Single Cycle Processor Design

ELT 3047

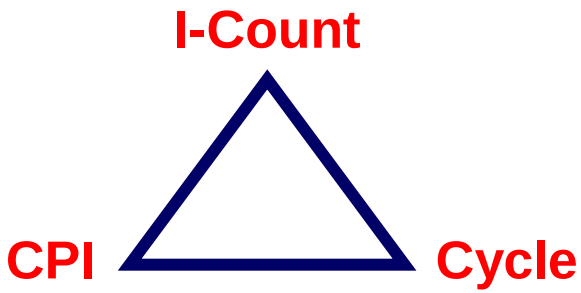
Computer Architecture

Presentation Outline

- Designing a Processor: Step-by-Step
- Datapath Components and Clocking
- Assembling an Adequate Datapath
- Controlling the Execution of Instructions
- The Main Controller and ALU Controller
- Drawback of the single-cycle processor design

The Performance Perspective

- Recall, performance is determined by:
  - Instruction count
  - Clock cycles per instruction (CPI)
  - Clock cycle time



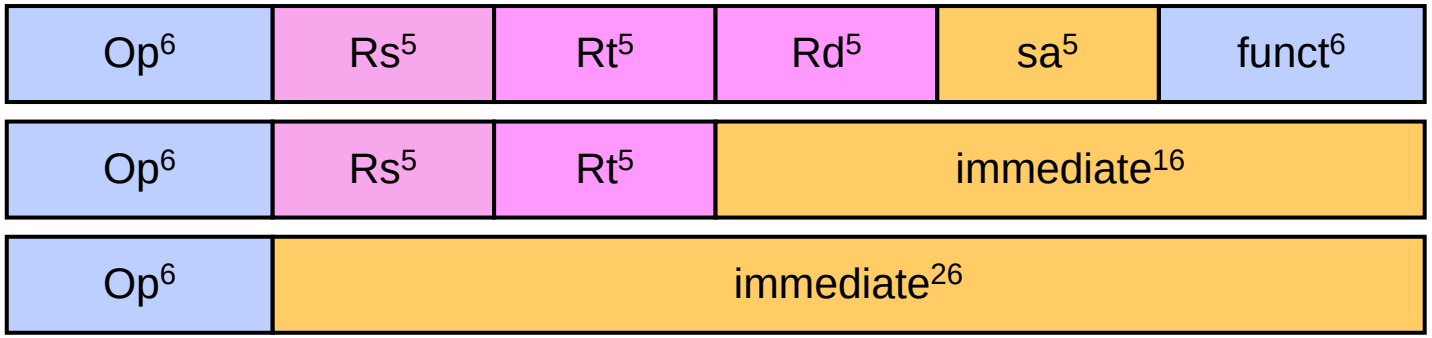
- Processor design will affect
  - Clock cycles per instruction
  - Clock cycle time
- Single cycle datapath and control design:
  - Advantage: One clock cycle per instruction
  - Disadvantage: long cycle time

Designing a Processor: Step-by-Step

- Analyze instruction set => datapath requirements
  - The meaning of each instruction is given by the register transfers
  - Datapath must include storage elements for ISA registers
  - Datapath must support each register transfer
- Select datapath components and clocking methodology
- Assemble datapath meeting the requirements
- Analyze implementation of each instruction
  - Determine the setting of control signals for register transfer
- Assemble the control logic

Review of MIPS Instruction Formats

- All instructions are 32-bit wide
- Three instruction formats: R-type, I-type, and J-type



  - Op^6: 6-bit opcode of the instruction
  - Rs^5, Rt^5, Rd^5: 5-bit source and destination register numbers
  - sa^5: 5-bit shift amount used by shift instructions
  - $funct^6$: 6-bit function field for R-type instructions
  - $immediate^{16}$: 16-bit immediate value or address offset
  - $immediate^{26}$: 26-bit target address of the jump instruction

MIPS Subset of Instructions

- Only a subset of the MIPS instructions are considered
  - ALU instructions (R-type): **add, sub, and, or, xor, slt**
  - Immediate instructions (I-type): **addi, slti, andi, ori, xori**
  - Load and Store (I-type): **lw, sw**
  - Branch (I-type): **beq, bne**
  - Jump (J-type): **j**
- This subset does not include all the integer instructions
- But sufficient to illustrate design of datapath and control
- Concepts used to implement the MIPS subset are used to construct a broad spectrum of computers

Details of the MIPS Subset

Instruction		Meaning	Format					
add	rd, rs, rt	addition	op^6 = 0	rs^5	rt^5	rd^5	0	0x20
sub	rd, rs, rt	subtraction	op^6 = 0	rs^5	rt^5	rd^5	0	0x22
and	rd, rs, rt	bitwise and	op^6 = 0	rs^5	rt^5	rd^5	0	0x24
or	rd, rs, rt	bitwise or	op^6 = 0	rs^5	rt^5	rd^5	0	0x25
xor	rd, rs, rt	exclusive or	op^6 = 0	rs^5	rt^5	rd^5	0	0x26
slt	rd, rs, rt	set on less than	op^6 = 0	rs^5	rt^5	rd^5	0	0x2a
addi	rt, rs, im^{16}	add immediate	0x08	rs^5	rt^5	im^{16}		
slti	rt, rs, im^{16}	slt immediate	0x0a	rs^5	rt^5	im^{16}		
andi	rt, rs, im^{16}	and immediate	0x0c	rs^5	rt^5	im^{16}		
ori	rt, rs, im^{16}	or immediate	0x0d	rs^5	rt^5	im^{16}		
xori	rt, im^{16}	xor immediate	0x0e	rs^5	rt^5	im^{16}		
lw	rt, im^{16}(rs)	load word	0x23	rs^5	rt^5	im^{16}		
sw	rt, im^{16}(rs)	store word	0x2b	rs^5	rt^5	im^{16}		
beq	rs, rt, im^{16}	branch if equal	0x04	rs^5	rt^5	im^{16}		
bne	rs, rt, im^{16}	branch not equal	0x05	rs^5	rt^5	im^{16}		
j	im^{26}	jump	0x02	im^{26}				

Register Transfer Level (RTL)

- RTL is a description of data flow between registers
- RTL gives a meaning to the instructions
- All instructions are fetched from memory at address PC

Instruction	RTL Description	
ADD	Reg(Rd) ← Reg(Rs) + Reg(Rt);	PC ← PC + 4
SUB	Reg(Rd) ← Reg(Rs) – Reg(Rt);	PC ← PC + 4
ORI	Reg(Rt) ← Reg(Rs) | zero_ext(Im16);	PC ← PC + 4
LW	Reg(Rt) ← MEM[Reg(Rs) + sign_ext(Im16)];	PC ← PC + 4
SW	MEM[Reg(Rs) + sign_ext(Im16)] ← Reg(Rt);	PC ← PC + 4
BEQ	if (Reg(Rs) == Reg(Rt))	
	PC ← PC + 4 + 4 × sign_extend(Im16)	
	else PC ← PC + 4	

Instructions are Executed in Steps

- **R-type**

Fetch instruction:	Instruction ← MEM[PC]
Fetch operands:	data1 ← Reg(Rs), data2 ← Reg(Rt)
Execute operation:	ALU_result ← func(data1, data2)
Write ALU result:	Reg(Rd) ← ALU_result
Next PC address:	PC ← PC + 4

- **I-type**

Fetch instruction:	Instruction ← MEM[PC]
Fetch operands:	data1 ← Reg(Rs), data2 ← Extend(imm16)
Execute operation:	ALU_result ← op(data1, data2)
Write ALU result:	Reg(Rt) ← ALU_result
Next PC address:	PC ← PC + 4

- **BEQ**

Fetch instruction:	Instruction ← MEM[PC]
Fetch operands:	data1 ← Reg(Rs), data2 ← Reg(Rt)
Equality:	zero ← subtract(data1, data2)
Branch:	if (zero) PC ← PC + 4 + 4×sign_ext(imm16)
	else PC ← PC + 4

Instruction Execution – cont'd

- **LW**

Fetch instruction:	Instruction ← MEM[PC]
Fetch base register:	base ← Reg(Rs)
Calculate address:	address ← base + sign_extend(imm16)
Read memory:	data ← MEM[address]
Write register Rt:	Reg(Rt) ← data
Next PC address:	PC ← PC + 4

- **SW**

Fetch instruction:	Instruction ← MEM[PC]
Fetch registers:	base ← Reg(Rs), data ← Reg(Rt)
Calculate address:	address ← base + sign_extend(imm16)
Write memory:	MEM[address] ← data
Next PC address:	PC ← PC + 4

concatenation

- **Jump**

Fetch instruction:	Instruction ← MEM[PC]
Target PC address:	target ← PC[31:28] || Imm26 || '00'
Jump:	PC ← target

Requirements of the Instruction Set

- Memory
  - Instruction memory where instructions are stored
  - Data memory where data is stored
- Registers
  - 31 × 32-bit general purpose registers, R0 is always zero
  - Read source register Rs
  - Read source register Rt
  - Write destination register Rt or Rd
- Program counter PC register and Adder to increment PC
- Sign and Zero extender for immediate constant
- ALU for executing instructions

Next . . .

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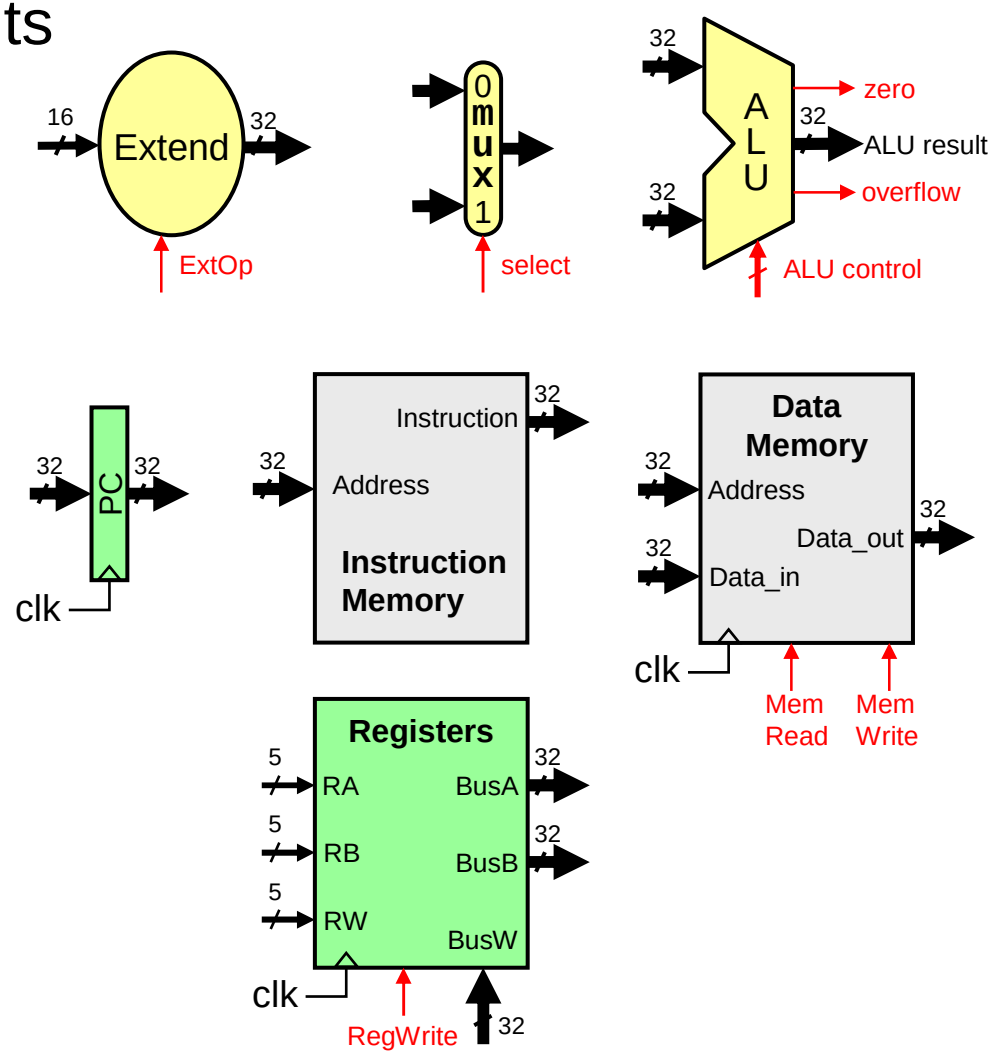
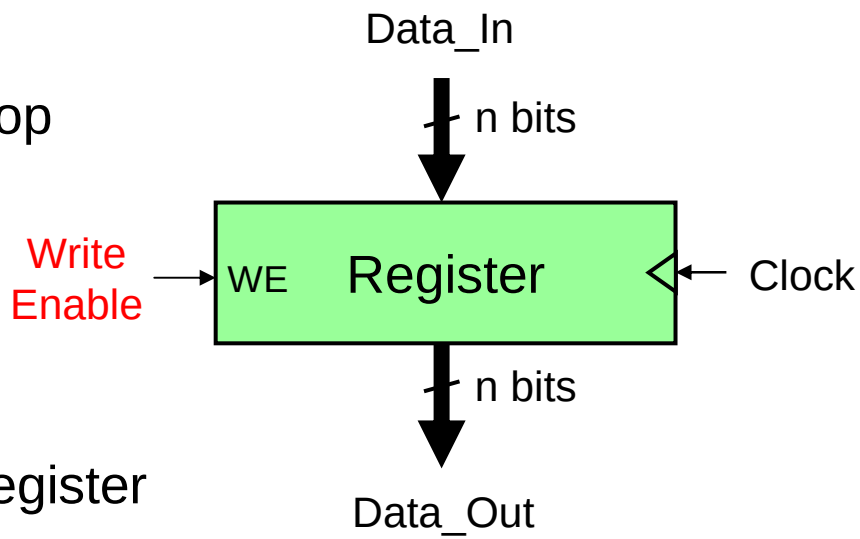
Components of the Datapath

- Combinational Elements
  - ALU, Adder
  - Immediate extender
  - Multiplexers
- Storage Elements
  - Instruction memory
  - Data memory
  - PC register
  - Register file
- Clocking methodology
  - Timing of writes



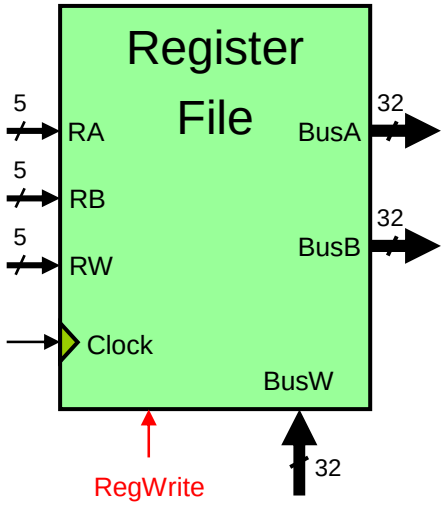
Register Element

- Register
  - Similar to the D-type Flip-Flop
- n-bit input and output
- Write Enable (WE):
  - Enable / disable writing of register
  - Negated (0): Data_Out will not change
  - Asserted (1): Data_Out will become Data_In after clock edge
- Edge triggered Clocking
  - Register output is modified at clock edge

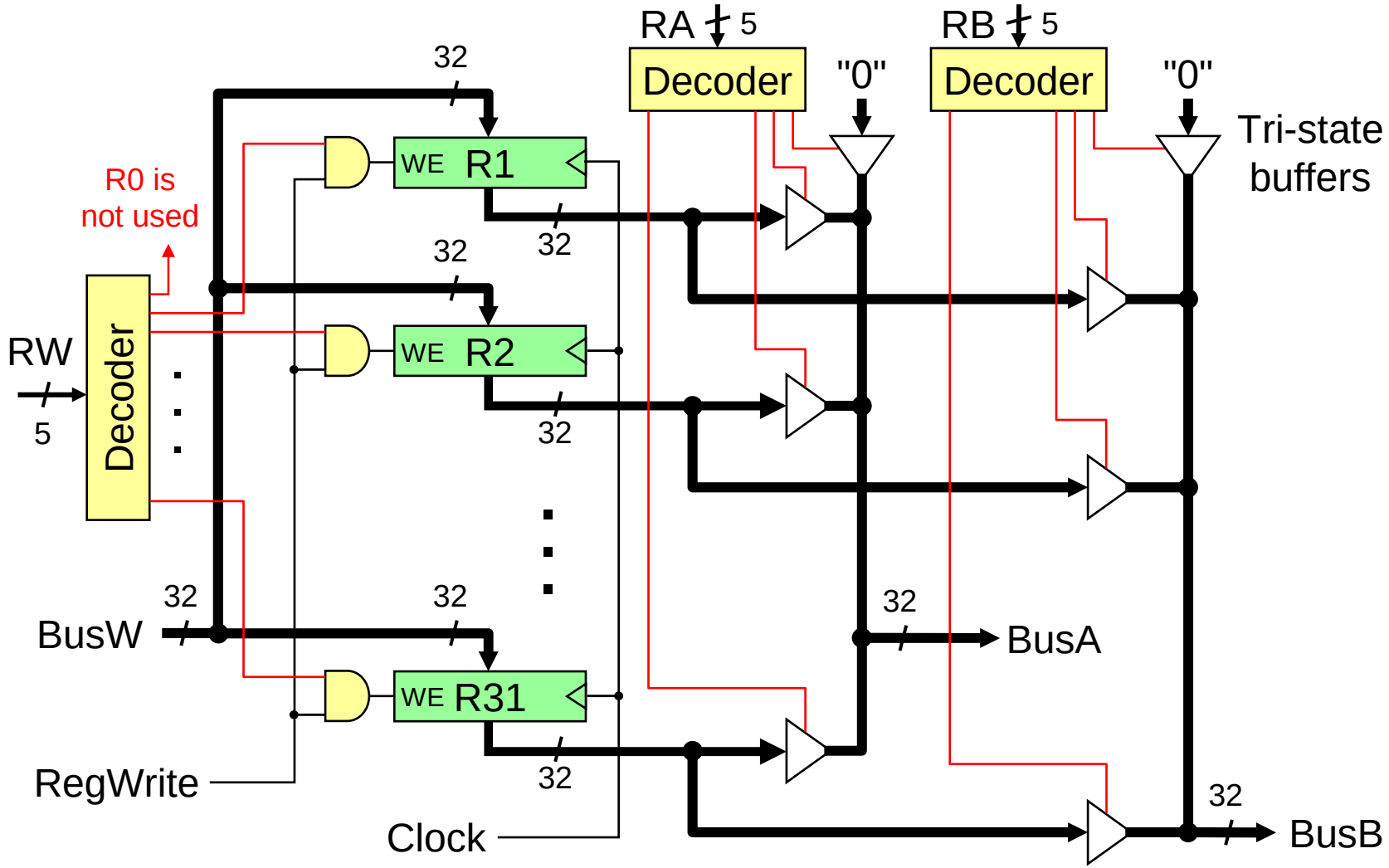


MIPS Register File

- Register File consists of 32 × 32-bit registers
  - BusA and BusB: 32-bit output busses for reading 2 registers
  - BusW: 32-bit input bus for writing a register when RegWrite is 1
  - Two registers read and one written in a cycle
- Registers are selected by:
  - RA selects register to be read on BusA
  - RB selects register to be read on BusB
  - RW selects the register to be written
- Clock input
  - The clock input is used ONLY during write operation
  - During read, register file behaves as a combinational logic block
    - RA or RB valid => BusA or BusB valid after access time

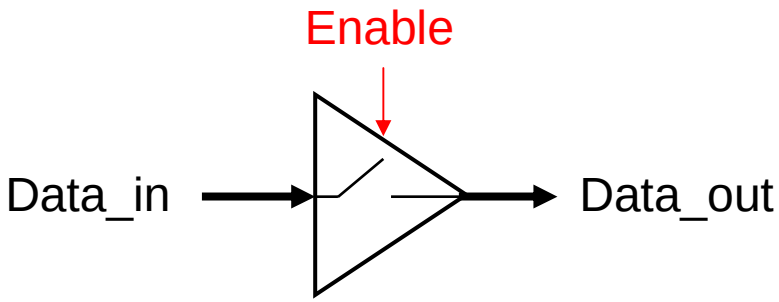


Details of the Register File



Tri-State Buffers

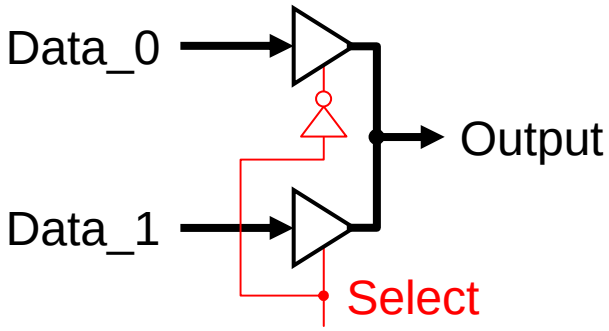
- Allow multiple sources to drive a single bus
- Two Inputs:
  - Data_in
  - Enable (to enable output)



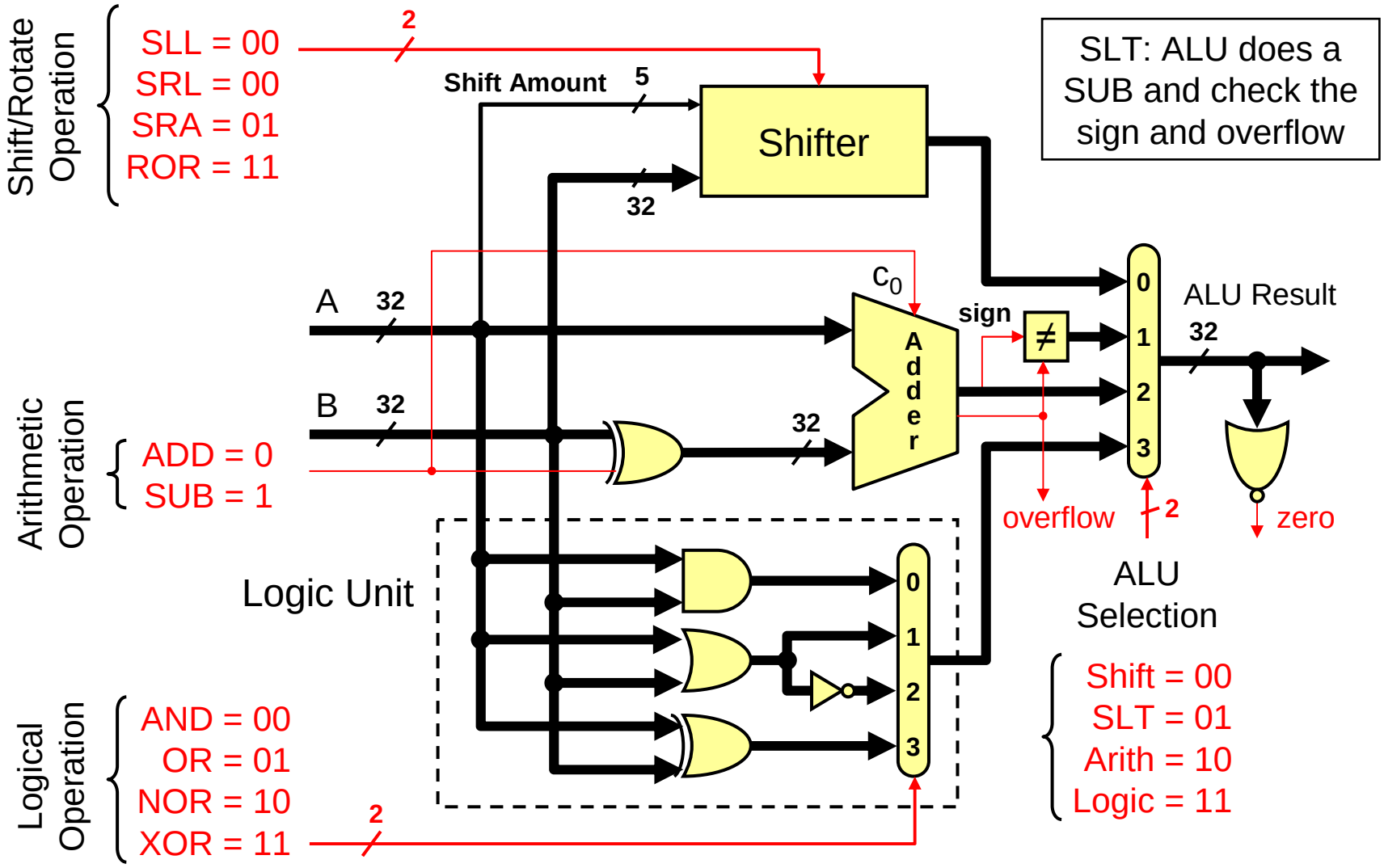
- One Output: Data_out
  - If (Enable) Data_out = Data_in

    else Data_out = High Impedance state (output is disconnected)

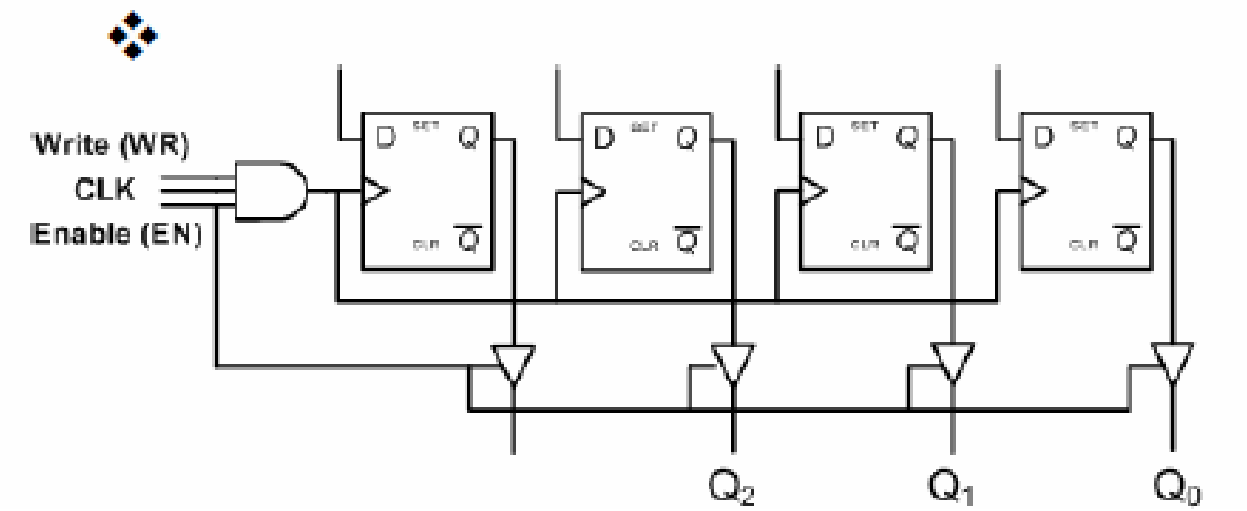
- Tri-state buffers can be used to build multiplexors

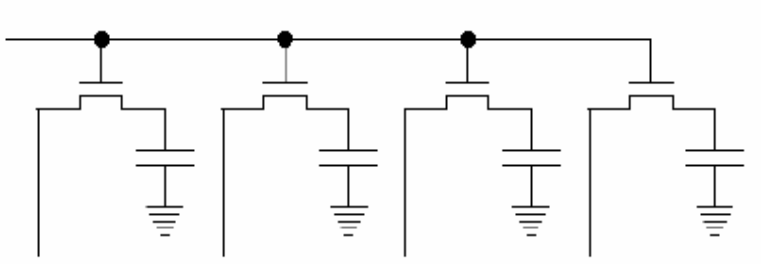


Building a Multifunction ALU



- Lưu trữ trong thanh ghi hoặc từ nhớ





Phép toán dịch

- **Dịch logic**
  - Các chữ số trống được điền bằng 0
  - Sang phải1 bit: $srl_1(a_{n-1},a_{n-2},\ldots,a_0) = (0,a_{n-1},a_{n-2},\ldots,a_1)$
  - Sang trái 1 bit: $sll_1(a_{n-1},a_{n-2},\ldots,a_0) = (a_{n-2},a_{n-3},\ldots,a_0,0)$
  - Dùng để triển khai bộ nhân và chia không dấu

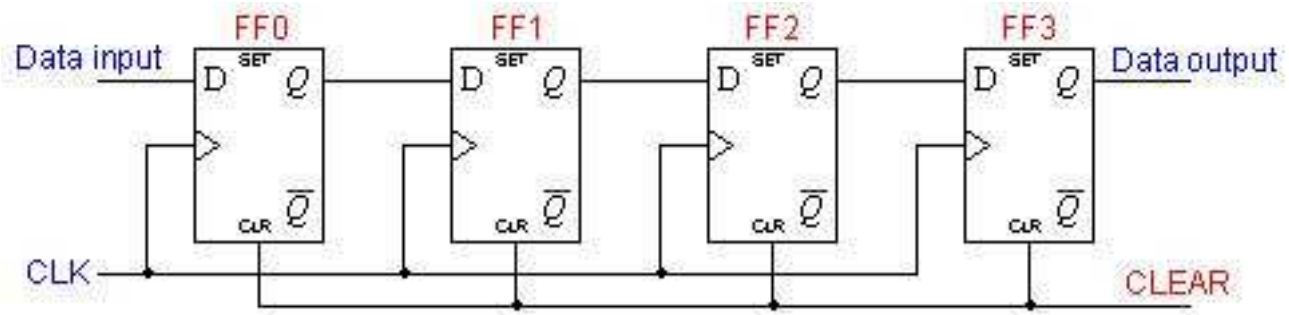
0	a_{n-2}	...	a_1	a_0
a_{n-1}	a_{n-2}	...	a_1	a_0

- **Dịch số học**
  - Bít dấu (MSB) không được thay đổi
    - dịch phải sao chép bít dấu vào các chữ số trống
    - dịch trái không dịch bít dấu
  - Sang phải 1 bit: $sra_1(a_{n-1},a_{n-2},\ldots,a_0) = (a_{n-1},a_{n-1},a_{n-2},\ldots,a_1)$
  - Sang trái 1 bit: $sla_1(a_{n-1},a_{n-2},\ldots,a_0) = (a_{n-1},a_{n-3},\ldots,a_0,0)$
  - Dùng để triển khai bộ nhân và chia có dấu

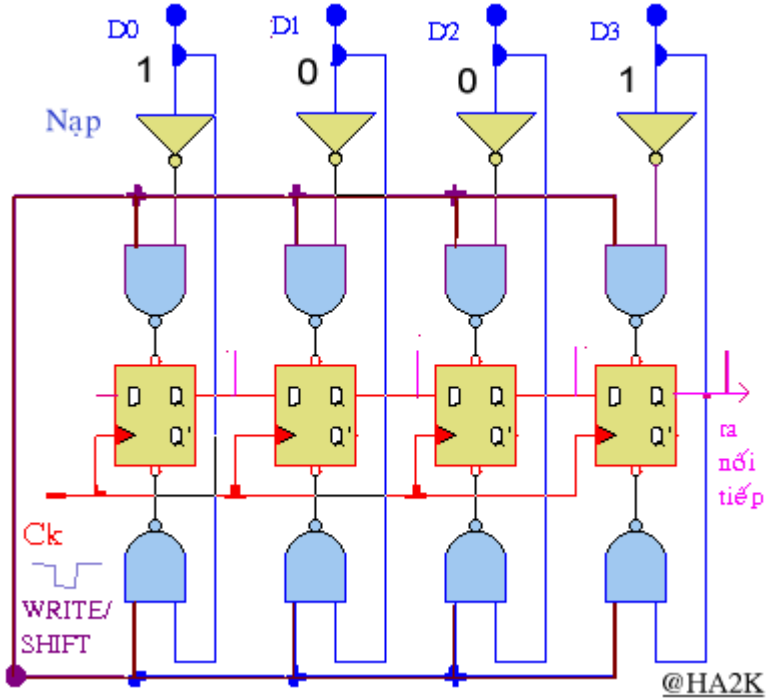
a_{n-1}	a_{n-2}	...	a_1	a_0
a_{n-1}	a_{n-2}	a_{n-3}	...	a_0

- Kết quả sai và xảy ra hiện tượng tràn nếu: $a_{n-1} \neq a_{n-2}$

Thanh ghi dịch vào/ra - nối tiếp 4 bit dùng 4 Flip-Flop D



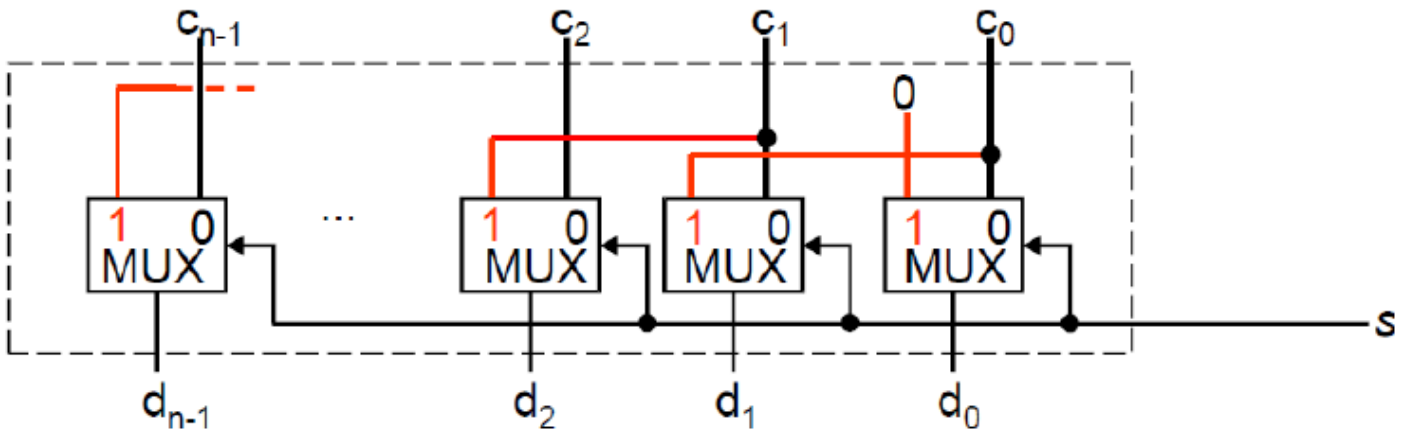
Thanh ghi dịch vào-song song / ra - nối tiếp 4 bit



CLEAR	Q0	Q1	Q2	Q3
	0	0	0	0

Bộ dịch

- Dịch trái 0 hoặc 1 bít

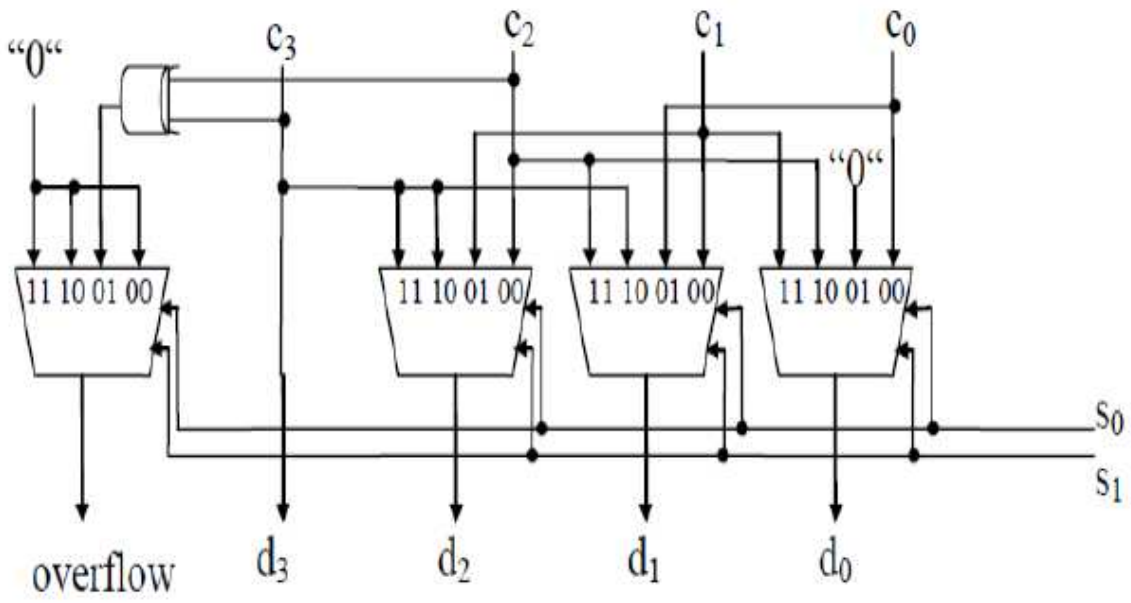


- Có thể thiết kế bộ dịch cả trái và phải

Bộ dịch

❖Bộ dịch trái 1 bít, và dịch phải 2 bít

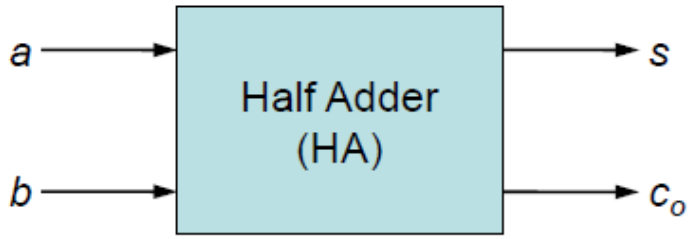
shift by	s_1	s_0
1	0	1
0	0	0
-1	1	1
-2	1	0



Bộ cộng nửa, cộng 2 số 1 bit

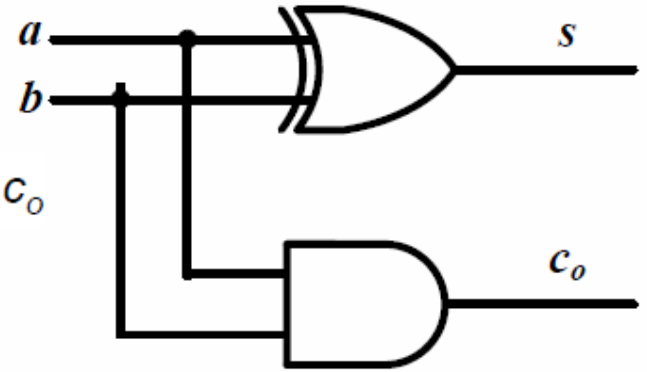
Tín hiệu vào: *a*, *b*

Tín hiệu ra: *s* (sum), c_o (carry out)



a	*b*	*s*	c_o
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Xác định biểu thức Bool cho *s*, và c_o



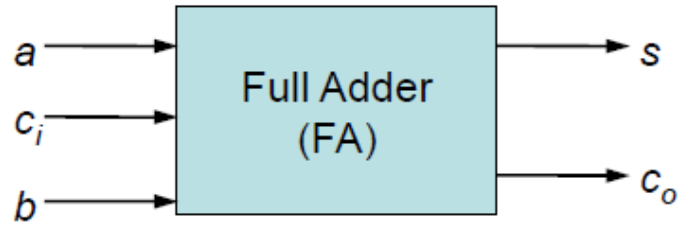
Bộ cộng đầy đủ, cộng 3 số 1 bit

Tín hiệu vào: a, b, c_i (carry in)

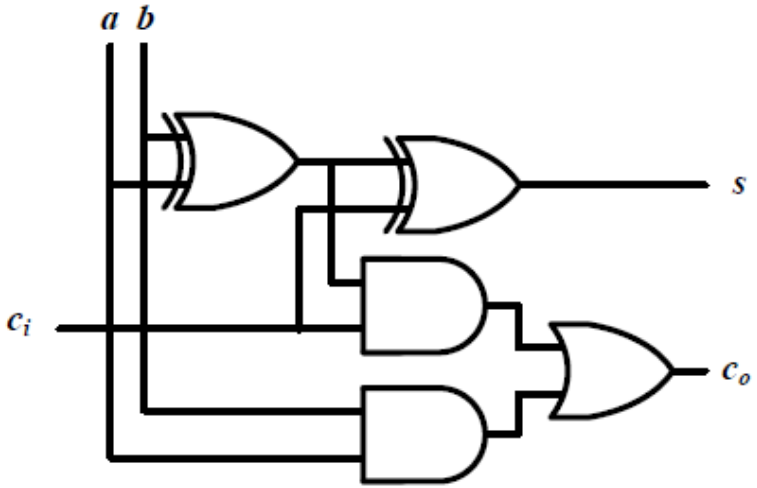
Tín hiệu ra: s (sum), c_o (carry out)

Câu hỏi:

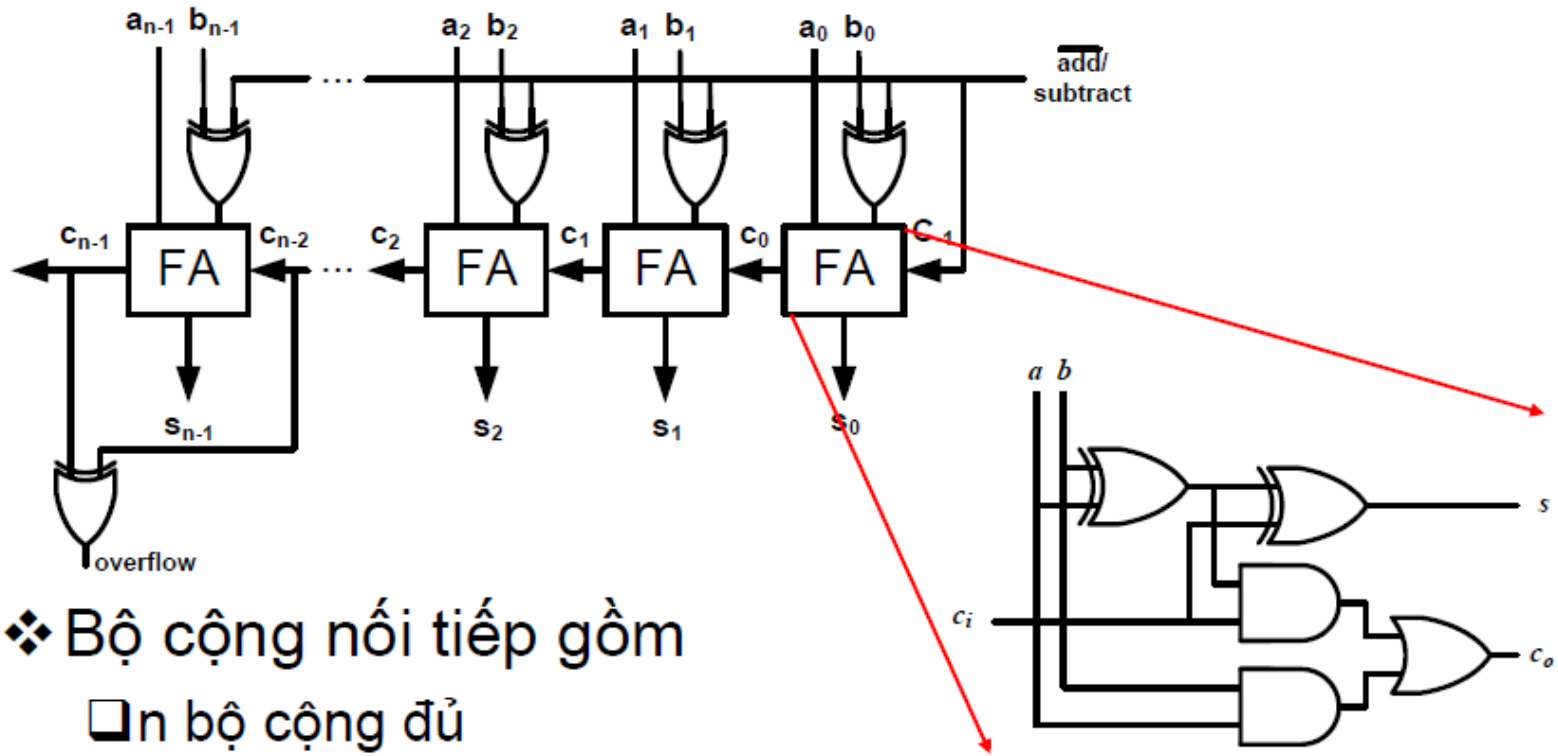
Xác định biểu thức Bool cho s, và c_o



a	b	c_i	s	c_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Bộ cộng 2 số n bít dạng bù 2



- Bộ cộng nối tiếp gồm
  - n bộ cộng đủ
  - n cổng logic xor để tính số đảo khi trừ
  - Cổng logic xor để phát hiện tràn

Tốc độ bộ cộng

- Bộ cộng nối tiếp
  - Tín hiệu nhớ lan truyền (*ripples*) qua tất cả các bộ cộng "*ripple carry adder*"
  - Độ trễ tăng tuyến tính với số lượng bộ cộng (số bit của mỗi toán hạng)
    - ✓ Bít nhớ: $t_{ar}(c_n) = t_{ar}(c_{n-1}) + 2 = 3 + 2*(n-1)$
    - ✓ Bít tổng: $t_{ar}(s_n) = t_{ar}(c_n) + 1 = 4 + 2*(n-1)$
- Tăng tốc bằng bộ cộng tính bit nhớ trước (*Carry lookahead Adder*)

Bộ cộng CLA

- Với Ripple-Carry Adder, bit nhớ được tính dựa trên các bít nhớ trước đó ➔ tốc độ chậm
- Tăng tốc độ, tính bit nhớ ở mỗi cột trực tiếp từ tín hiệu đầu vào

$$s_i = a_i \oplus b_i \oplus c_{i-1}$$

$$\begin{aligned} c_i &= a_i b_i + a_i c_{i-1} + b_i c_{i-1} \\ &= a_i b_i + c_{i-1}(a_i + b_i) \\ &= a_i b_i + c_{i-1}(a_i \oplus b_i) \end{aligned}$$

Tín hiệu tạo nhớ: $g_i = a_i b_i$
Tạo ra c_i khi $a_i = b_i = 1$

Lan truyền nhớ: $p_i = (a_i \oplus b_i)$
Truyền nhớ từ đầu vào đến đầu ra khi $a_i \oplus b_i = 1$

- Bit nhớ đầu ra của cột *i* được tính từ tín hiệu tạo nhớ và tín hiệu lan truyền nhớ $c_i = g_i + c_{i-1}\, p_i$

Bộ cộng CLA

- Tính toán bit nhớ

$$c_0 = g_0 + p_0 c_{-1}$$

$$c_1 = g_1 + p_1 c_0 = g_1 + p_1 g_0 + p_1 p_0 c_{-1}$$

$$c_2 = g_2 + p_2 c_1 = g_2 + p_2 g_1 + p_2 p_1 g_0 + p_2 p_1 p_0 c_{-1}$$

$$c_3 = g_3 + p_3 c_2 = g_3 + p_3 g_2 + p_3 p_2 g_1 + p_3 p_2 p_1 g_0 + p_3 p_2 p_1 p_0 c_{-1}$$

- Mỗi công thức nhớ trên có thể được triển bởi một mạch logic 2 mức
- Để tính toán p_i, g_i ta cần mạch logic 1 mức từ đầu vào
- Vậy cần tối đa 3 mức từ đầu vào để tính được tín hiệu nhớ

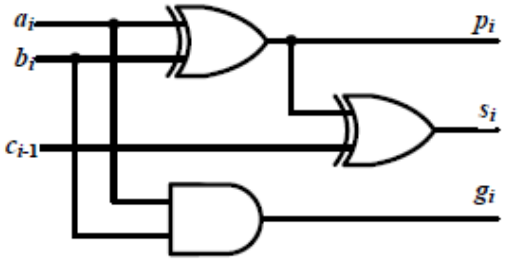
➔Tăng tốc độ

➔Cần cổng AND n+2 đầu vào cho c_n!

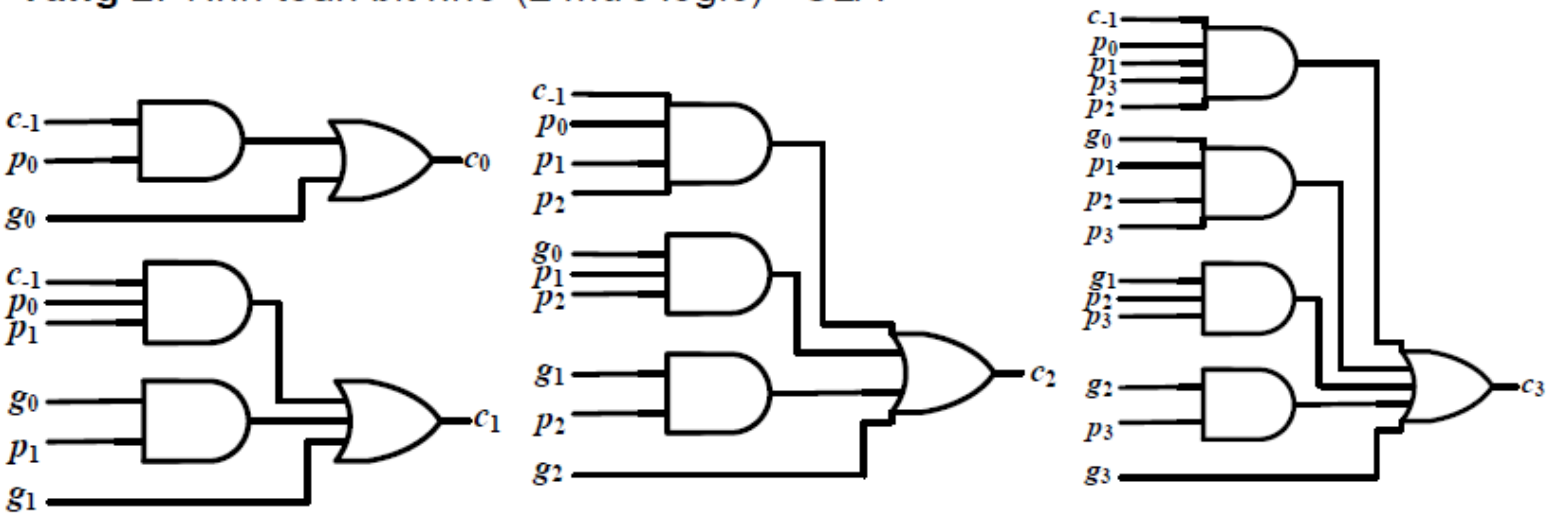
Bộ cộng CLA

Gồm 2 tầng

Tầng 1: Tính toán tổng, tính hiệu tạo nhớ và truyền nhớ (1 mức logic) - PFA



Tầng 2: Tính toán bit nhớ (2 mức logic) - CLA



Phép nhân không dấu

- Nhân lần lượt các cột của số bị nhân và số nhân được tích cục bộ
- Các tích cục bộ được cộng với nhau theo cột

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
					a_3b_0	a_2b_0	a_1b_0	a_0b_0
+				a_3b_1	a_2b_1	a_1b_1	a_0b_1	
+			a_3b_2	a_2b_2	a_1b_2	a_0b_2		
+		a_3b_3	a_2b_3	a_1b_3	a_0b_3			
	s_7	s_6	s_5	s_4	s_3	s_2	s_1	s_0

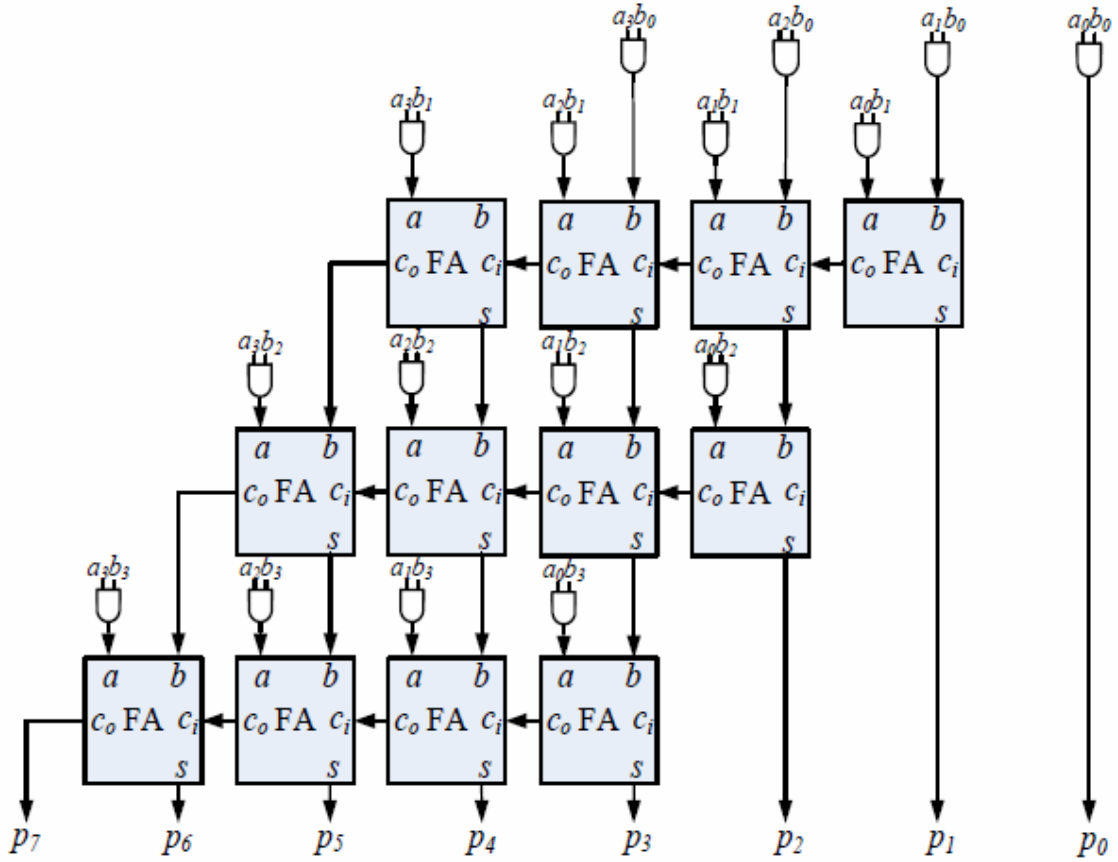
a	b	a*b
0	0	0
0	1	0
1	0	0
1	1	1

- Ví dụ

1	0	1	1	*	0	0	1	1	11*3
					1	0	1	1	
+				1	0	1	1		
+			0	0	0	0			
+		0	0	0	0				
	0	0	1	0	0	0	0	1	33

Bộ nhân không dấu song song

- Sử dụng logic tổ hợp



Phép nhân có dấu

- Mở rộng bít dấu cho các tích cục bộ
- Với tích cục bộ của bit dấu số b_3, cần lấy số đối (số bù 2)

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
	a_3b_0	a_3b_0	a_3b_0	a_3b_0	a_3b_0	a_2b_0	a_1b_0	a_0b_0
+	a_3b_1	a_3b_1	a_3b_1	a_3b_1	a_2b_1	a_1b_1	a_0b_1	
+	a_3b_2	a_3b_2	a_3b_2	a_2b_2	a_1b_2	a_0b_2		
+	$\overline{a_3b_3}$	$\overline{a_3b_3}$	$\overline{a_2b_3}$	$\overline{a_1b_3}$	$\overline{a_0b_3}$			
+					1			
	p_7	p_6	p_5	p_4	p_3	p_2	p_1	p_0

Ví dụ 2.9 – Phép nhân có dấu

1	0	1	1	*	0	0	1	1	-5*3
	1	1	1	1	1	0	1	1	
+	1	1	1	1	0	1	1		
+	0	0	0	0	0	0			
+	1	1	1	1	1				
					1				
						1			
					1				
			1	0					
			1						
	1	0							
	1								
	1	1	1	1	0	0	0	1	-15

Phép nhân có dấu

❖ Đơn giản hóa

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
	a_3b_0	a_3b_0	a_3b_0	a_3b_0	a_3b_0	a_2b_0	a_1b_0	a_0b_0
+	a_3b_1	a_3b_1	a_3b_1	a_3b_1	a_2b_1	a_1b_1	a_0b_1	
+	a_3b_2	a_3b_2	a_3b_2	a_2b_2	a_1b_2	a_0b_2		
+	$\overline{a_3b_3}$	$\overline{a_3b_3}$	$\overline{a_2b_3}$	$\overline{a_1b_3}$	$\overline{a_0b_3}$			
					1			
	p_7	p_6	p_5	p_4	p_3	p_2	p_1	p_0

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
						a_2b_0	a_1b_0	a_0b_0
+					a_2b_1	a_1b_1	a_0b_1	
+				a_2b_2	a_1b_2	a_0b_2		
+			$\overline{a_2b_3}$	$\overline{a_1b_3}$	$\overline{a_0b_3}$			
+					1			
-					a_3b_0			
-				a_3b_1				
-			a_3b_2					
-		$\overline{a_3b_3}$						
	p_7	p_6	p_5	p_4	p_3	p_2	p_1	p_0

Phép nhân có dấu

❖ Đơn giản hóa

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
						a_2b_0	a_1b_0	a_0b_0
+					a_2b_1	a_1b_1	a_0b_1	
+				a_2b_2	a_1b_2	a_0b_2		
+			$\overline{a_2b_3}$	$\overline{a_1b_3}$	$\overline{a_0b_3}$			
+					1			
+		$\overline{a_3b_3}$	a_3b_2	a_3b_1	a_3b_0			
	p_7	p_6	p_5	p_4	p_3	p_2	p_1	p_0

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
						a_2b_0	a_1b_0	a_0b_0
+					a_2b_1	a_1b_1	a_0b_1	
+				a_2b_2	a_1b_2	a_0b_2		
+			$\overline{a_2b_3}$	$\overline{a_1b_3}$	$\overline{a_0b_3}$			
+					1			
-					a_3b_0			
-				a_3b_1				
-			a_3b_2					
-		$\overline{a_3b_3}$						
	p_7	p_6	p_5	p_4	p_3	p_2	p_1	p_0

Phép nhân có dấu

❖ Đơn giản hóa

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
						a_2b_0	a_1b_0	a_0b_0
+					a_2b_1	a_1b_1	a_0b_1	
+				a_2b_2	a_1b_2	a_0b_2		
+			$\overline{a_2b_3}$	$\overline{a_1b_3}$	$\overline{a_0b_3}$			
+					1			
-	0	$\overline{a_3b_3}$	a_3b_2	a_3b_1	a_3b_0			
	p_7	p_6	p_5	p_4	p_3	p_2	p_1	p_0

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
						a_2b_0	a_1b_0	a_0b_0
+					a_2b_1	a_1b_1	a_0b_1	
+				a_2b_2	a_1b_2	a_0b_2		
+			$\overline{a_2b_3}$	$\overline{a_1b_3}$	$\overline{a_0b_3}$			
+					1			
+	1	a_3b_3	$\overline{a_3b_2}$	$\overline{a_3b_1}$	$\overline{a_3b_0}$			
					1			
	p_7	p_6	p_5	p_4	p_3	p_2	p_1	p_0

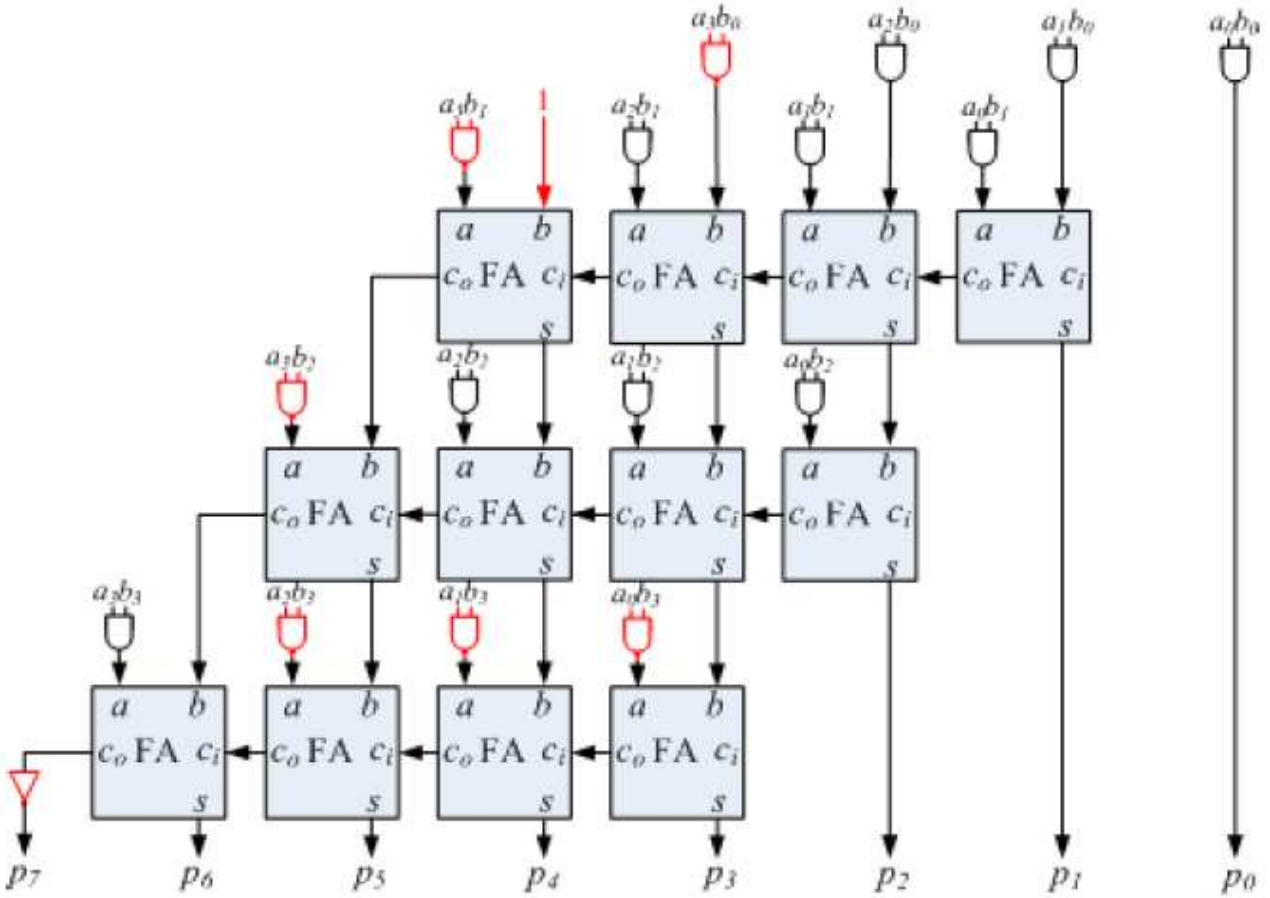
Phép nhân có dấu

❖ Đơn giản hóa

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
						a_2b_0	a_1b_0	a_0b_0
+					a_2b_1	a_1b_1	a_0b_1	
+				a_2b_2	a_1b_2	a_0b_2		
+			a_2b_3	a_1b_3	a_0b_3			
+					1			
+	1	a_3b_3	a_3b_2	a_3b_1	a_3b_0			
					1			
	p_7	p_6	p_5	p_4	p_3	p_2	p_1	P_0

a_3	a_2	a_1	a_0	*	b_3	b_2	b_1	b_0
					a_3b_0	a_2b_0	a_1b_0	a_0b_0
+				a_3b_1	a_2b_1	a_1b_1	a_0b_1	
+			a_3b_2	a_2b_2	a_1b_2	a_0b_2		
+		a_3b_3	a_2b_3	a_1b_3	a_0b_3			
+	1			1				
	p_7	p_6	p_5	p_4	p_3	p_2	p_1	p_0

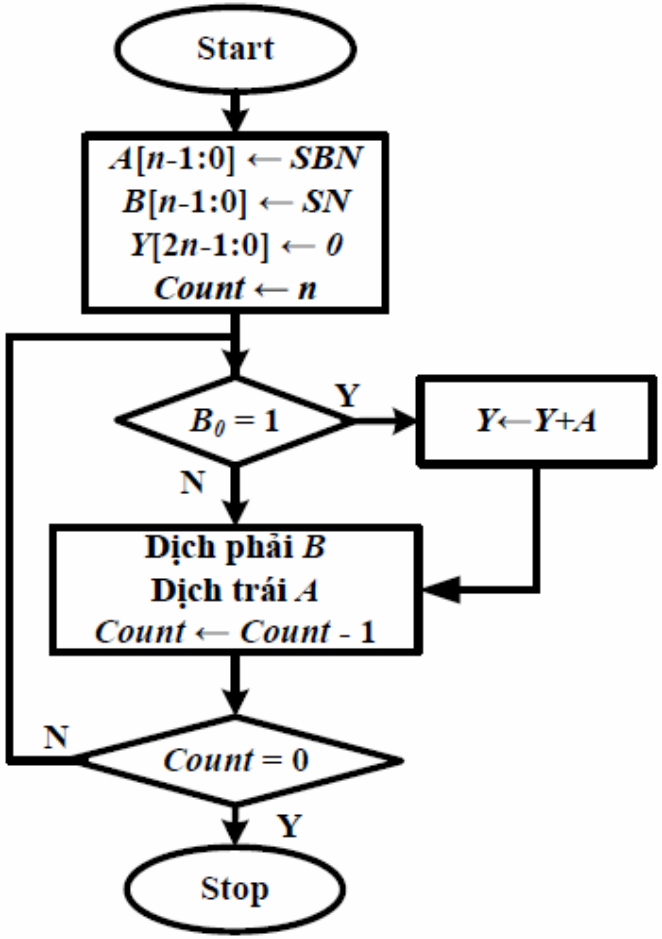
Bộ nhân có dấu



Bộ nhân nối tiếp

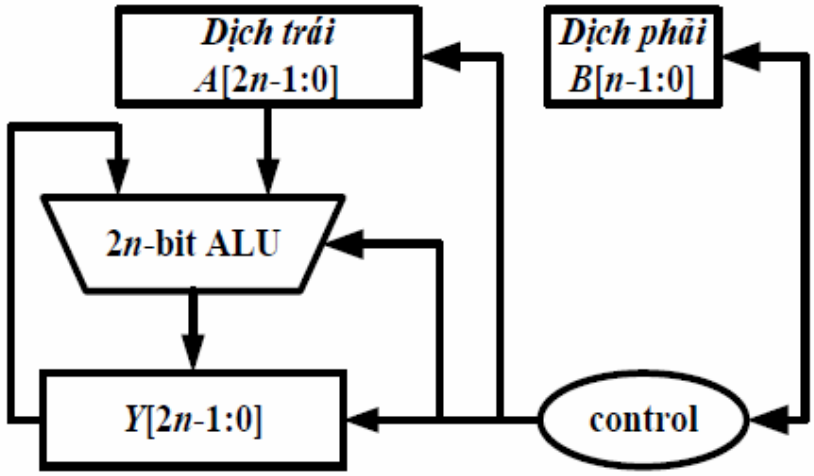
- Sử dụng bộ cộng để cộng các tích cục bộ
- Thực hiện phép nhân trong vài chu kỳ đồng hồ
- Lưu số bị nhân, số nhân và kết quả tạm thời trong các thanh ghi
- Với mỗi bit b_i của số nhân B (từ phải qua trái)
  - Nhân b_i với số bị nhân A và cộng tích với kết quả tổng tạm thời Y ⇔ Nếu $b_i = 1$ thì cộng A vào Y
  - Dịch A sang trái 1 bit

Bộ nhân nối tiếp



Triển khai gồm:

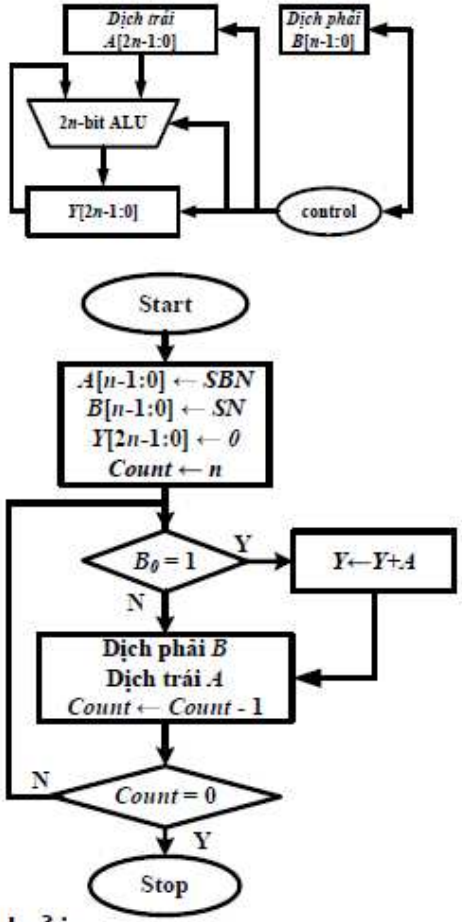
- 2 thanh ghi 2n bit
- 1 thanh ghi n bit
- 1 bộ cộng 2n bit
- 1 khối điều khiển



Ví dụ - Bộ nhân nối tiếp

Y	0	0	0	0	0	0	0	0
A	0	0	0	0	1	1	0	1
Y	0	0	0	0	1	1	0	1
A	0	0	0	1	1	0	1	0
Y	0	0	1	0	0	1	1	1
A	0	0	1	1	0	1	0	0
Y	0	0	1	0	0	1	1	1
A	0	1	1	0	1	0	0	0
Y	1	0	0	0	1	1	1	1
A	1	1	0	1	0	0	0	0

B	1	0	1	1	*Counter=4*
B	0	1	0	1	*Counter=3*
B	0	0	1	0	*Counter=2*
B	0	0	0	1	*Counter=1*
B	0	0	0	0	*Counter=0*

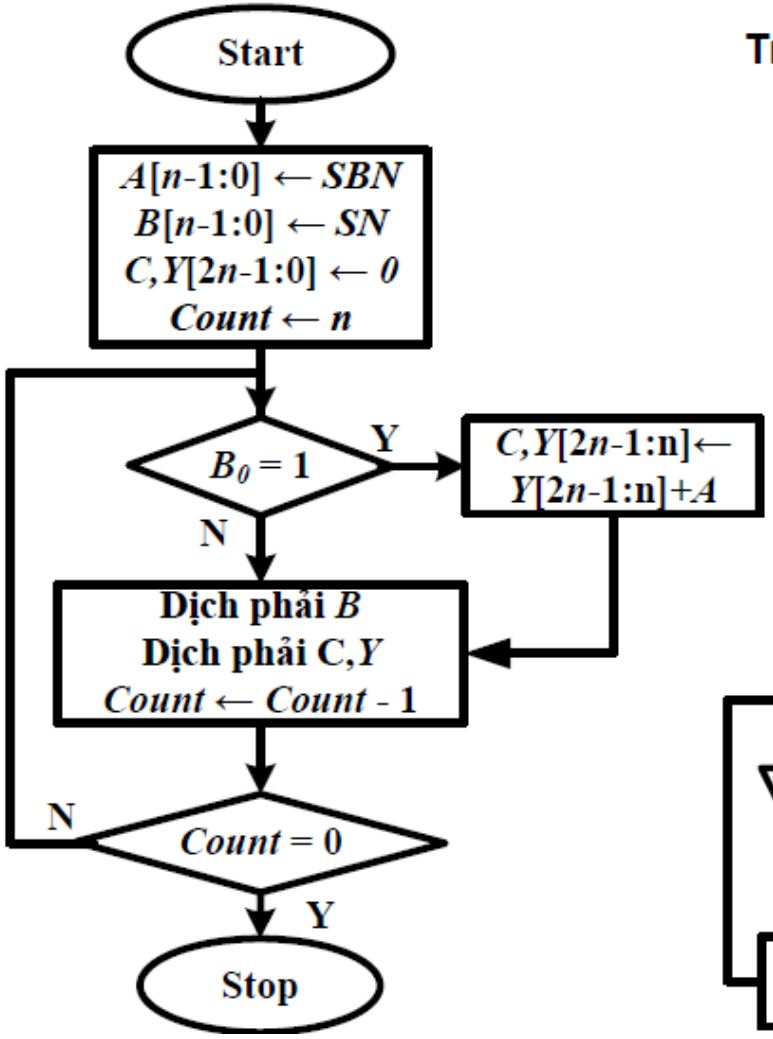


Nhận xét:

- ✓ Một nửa số bit của A luôn bằng 0
- ✓ Khi A dịch trái, bit 0 được thêm vào bên phải
- ➔ các bit LSB của tích không bị ảnh hưởng

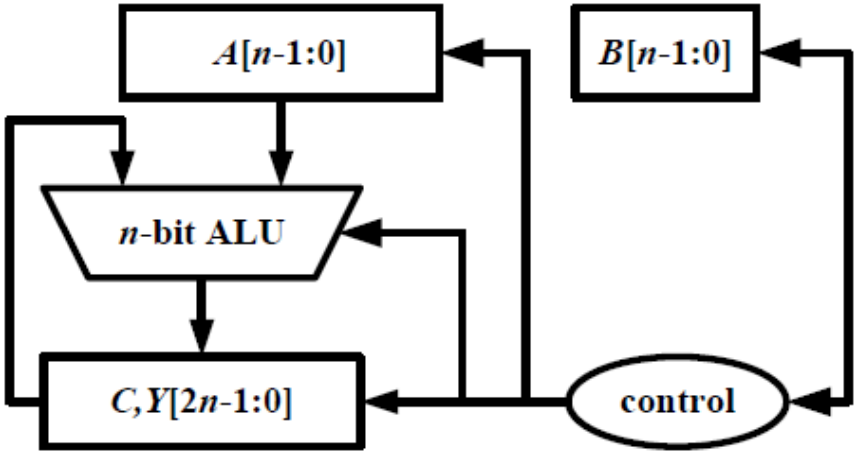
Ý tưởng: Giữ A ở phía trái của tích và dịch tích sang phải

Bộ nhân nối tiếp – Dùng *n*-bit ALU



Triển khai gồm:

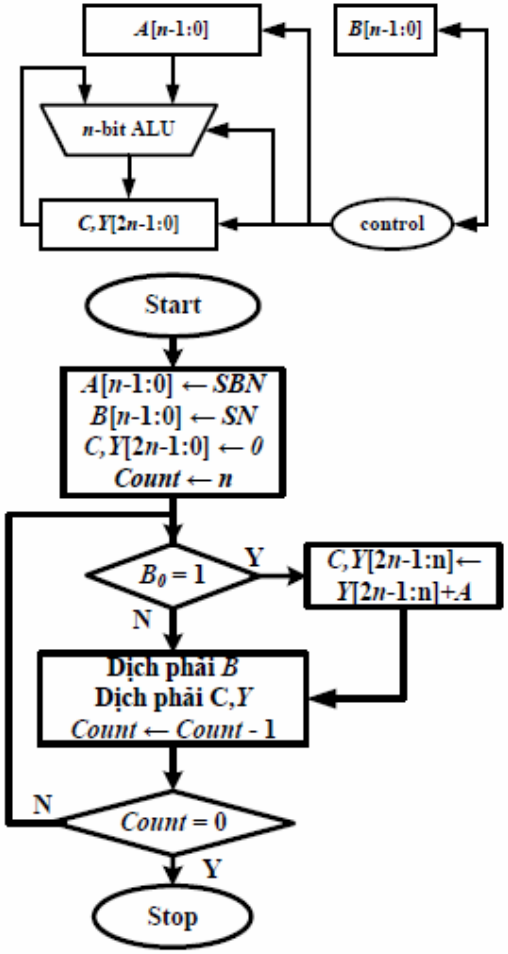
- 2 thanh ghi n bit
- 1 thanh ghi 2n+1 bit
- 1 bộ cộng n bit
- 1 khối điều khiển



Ví dụ – Bộ nhân nối tiếp

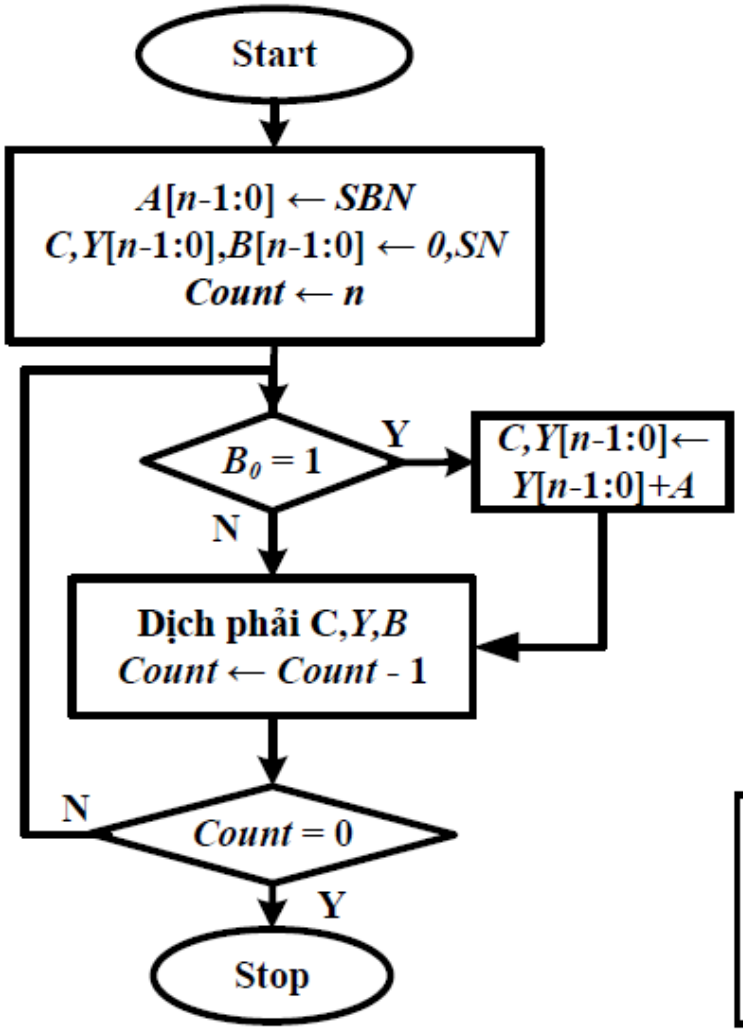
Y	0	0	0	0	0	0	0	0	0
A		1	1	0	1				
Y	0	1	1	0	1	0	0	0	0
Y	0	0	1	1	0	1	0	0	0
A		1	1	0	1				
Y	1	0	0	1	1	1	0	0	0
Y	0	1	0	0	1	1	1	0	0
A		1	1	0	1				
Y	0	1	0	0	1	1	1	0	0
Y	0	0	1	0	0	1	1	1	0
A		1	1	0	1				
Y	1	0	0	0	1	1	1	1	0
Y	0	1	0	0	0	1	1	1	1

B	1	0	1	1	*Counter=4*
B	0	1	0	1	*Counter=3*
B	0	0	1	0	*Counter=2*
B	0	0	0	1	*Counter=1*
B	0	0	0	0	*Counter=0*



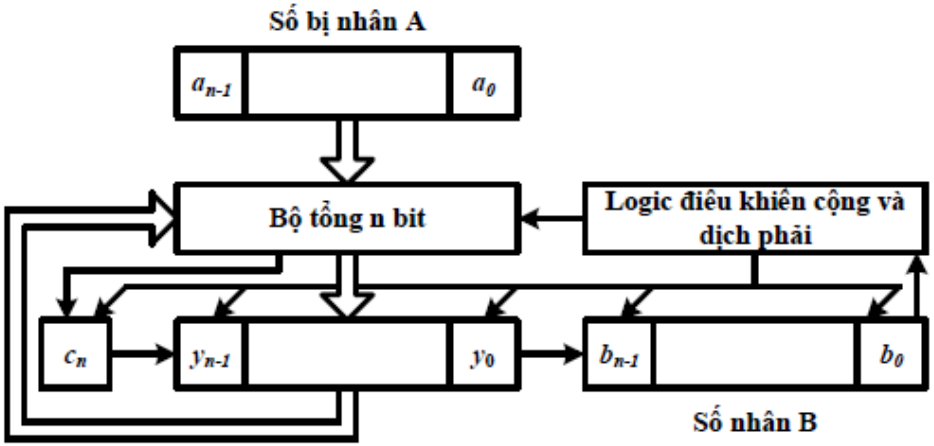
Nhận xét: Trong quá trình nhân chỉ một số bit của *Y* có ý nghĩa với kết quả

Bộ nhân nối tiếp



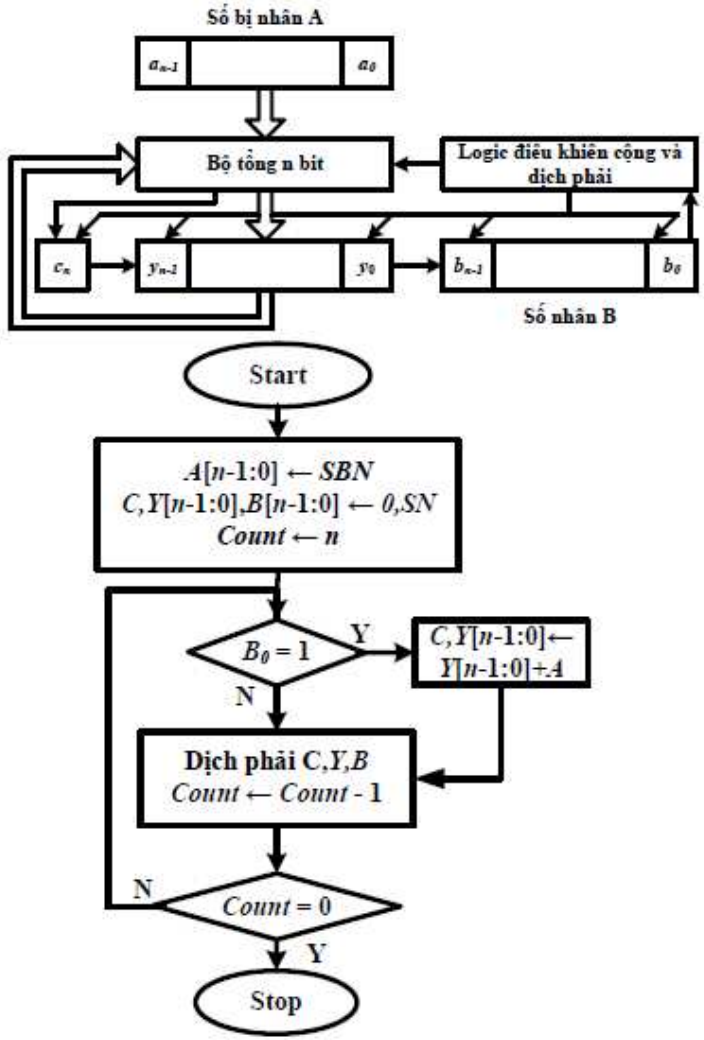
Triển khai gồm:

- 1 thanh ghi n bit
- 1 thanh ghi 2n+1 bit
- 1 bộ cộng n bit
- 1 khối điều khiển



Ví dụ – Bộ nhân nối tiếp

Y	0	0	0	0	0	1	0	1	1	*Counter=4*
A		1	1	0	1					
Y	0	1	1	0	1	1	0	1	1	
Y	0	0	1	1	0	1	1	0	1	*Counter=3*
A		1	1	0	1					
Y	1	0	0	1	1	1	1	0	1	
Y	0	1	0	0	1	1	1	1	0	*Counter=2*
A		1	1	0	1					
Y	0	1	0	0	1	1	1	1	0	
Y	0	0	1	0	0	1	1	1	1	*Counter=1*
A		1	1	0	1					
Y	1	0	0	0	1	1	1	1	1	
Y	0	1	0	0	0	1	1	1	1	*Counter=0*



Nhân Booth

Nhân với một chuỗi số 1

$$A * 1111 = A * (2^4 - 2^0) = A * 2^4 - A$$

➔ Dịch A sang trái 4 bit và trừ đi A

Số bị nhân B chứa chuỗi số 1 từ bit vị trí *v* đến bit vị trí *u*

$$(b_{n-1}, b_{n-2}, \ldots b_{u+1}, b_u, \ldots, b_v, b_{v-1}, .., b_0) = (b_{n-1}, b_{n-2}, \ldots 0, 1, \ldots, 1, 0, .., b_0)$$

Chuỗi bit có thể thay thế bằng $2^{u+1} - 2^v$

➔Các phép nhân và cộng cho các bít b_u đến b_v có thể được thay bằng phép dịch trái và phép trừ

Ví dụ:

B = 001110 (14_{10}), u = 3, v = 2 ➔ A x B = A* 2^4 – A*2^1

Ví dụ – Bộ nhân Booth

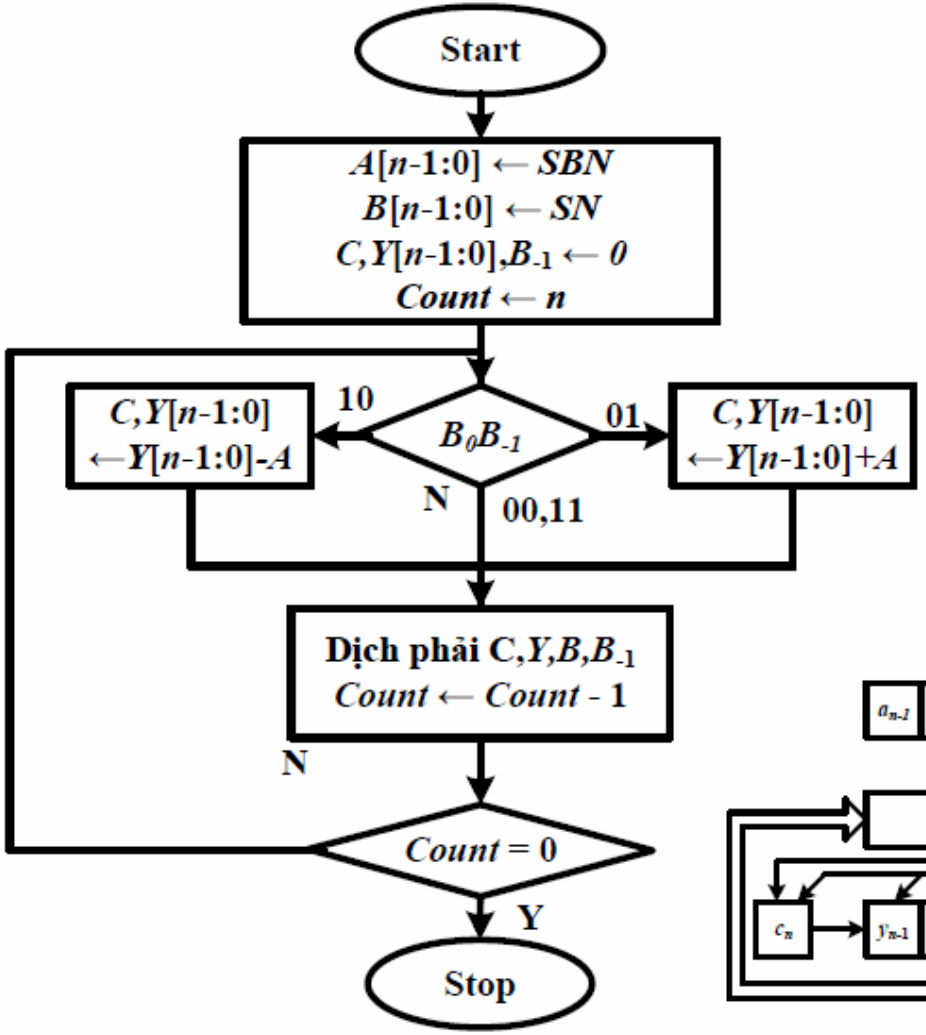
A	0	1	0	1	0	1
B	0	0	1	1	1	0
B	0	+1	0	0	-1	0

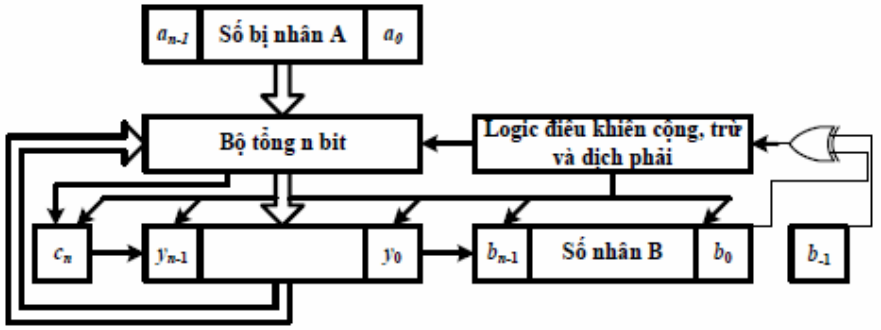
0	0	0	0	0	0	1	0	1	0	1	0
1	1	1	1	1	1	0	1	0	1	1	0
0	0	0	1	0	1	0	1	0	0	0	0
0	0	0	1	0	0	1	0	0	1	1	0

Thực hiện

1. Bắt đầu chuỗi số 1 (chuyển từ 0 sang 1, hai bit liên tiếp là 01): trừ đi số bị nhân
2. Trong chuỗi số 0, hoặc chuỗi số 1 (2 bit liên tiếp là 00 hoặc 11): dịch trái số bị nhân
3. Kết thúc chuỗi số 1 (chuyển từ 1 sang 0, hai bit liên tiếp là 10): cộng với số bị nhân

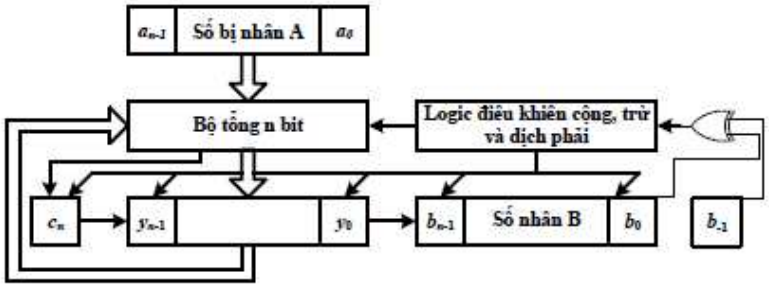
Thuật toán nhân Booth

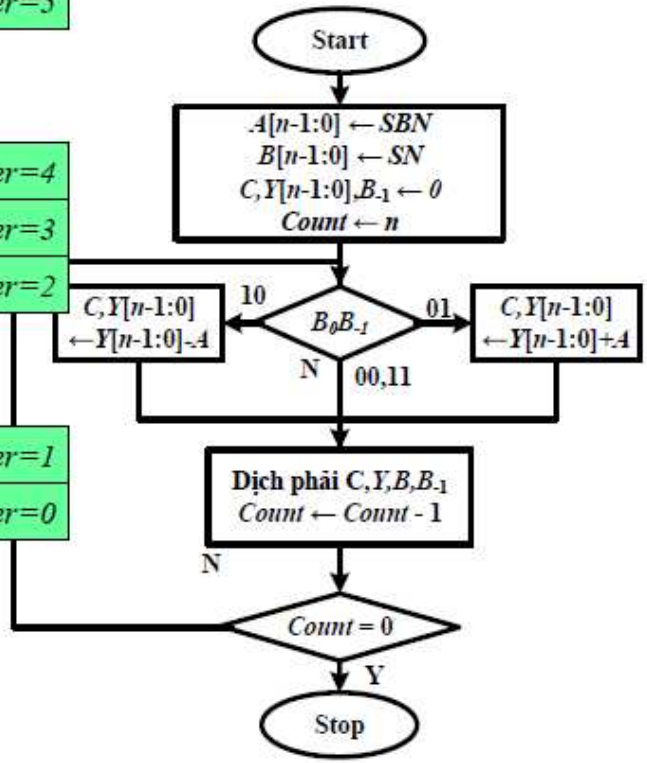




Ví dụ – Minh họa thuật toán Booth

A		0	1	0	1	0	1								
Y	0	0	0	0	0	0	0	0	0	1	1	1	0	0	*Counter=6*
Y	0	0	0	0	0	0	0	0	0	0	1	1	1	0	*Counter=5*
-A		1	0	1	0	1	1								
Y	1	1	0	1	0	1	1	0	0	0	1	1	1	0	
Y	1	1	1	0	1	0	1	1	0	0	0	1	1	1	*Counter=4*
Y	1	1	1	1	0	1	0	1	1	0	0	0	1	1	*Counter=3*
Y	1	1	1	1	1	0	1	0	1	1	0	0	0	1	*Counter=2*
+A		0	1	0	1	0	1								
Y	0	0	1	0	0	1	0	0	1	1	0	0	0	1	
Y	0	0	0	1	0	0	1	0	0	1	1	0	0	0	*Counter=1*
Y	0	0	0	0	1	0	0	1	0	0	1	1	0	0	*Counter=0*





Nhân Booth: Nhân có dấu

Vì a, b là 2 số có dấu dạng bù 2:

$$a = -2^{n-1}*a_n + 2^{n-2}*a_{n-2} + \ldots + 2*a_1 + a_0$$

Xét 2 bit liên tiếp (a_i, a_{i-1}), hiệu của chúng và hoạt động nhân:

a_i	a_{i-1}	$a_i - a_{i-1}$	action
1	0	-1	Trừ b, và dịch
0	1	1	Cộng b và dịch
0	0	0	Bỏ qua
1	1	1	Bỏ qua

Giá trị được tính toán bởi bộ nhân Booth:

$$(0 - a_0) * b + (a_0 - a_1) * b * 2 + (a_1 - a_2) * b * 2^2 \ldots +$$
$$(a_{n-3} - a_{n-2}) * b * 2^{n-2} + (a_{n-2} - a_{n-1}) * b * 2^{n-1}$$

Triển khai các số hạng và tối giản:

$$b * (-2^{n-1}*a_n + 2^{n-2}*a_{n-2} + \ldots + 2*a_1 + a_0) = b * a.$$

which is exactly the product of a and b.

Phép chia

Phép nhân

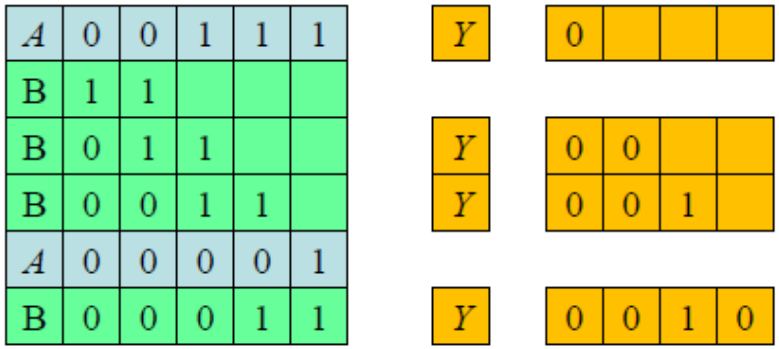
Cộng với số bị nhân và dịch trái số bị nhân

Tối ưu phần cứng: cộng với số bị nhân và dịch phải tích

Phép chia

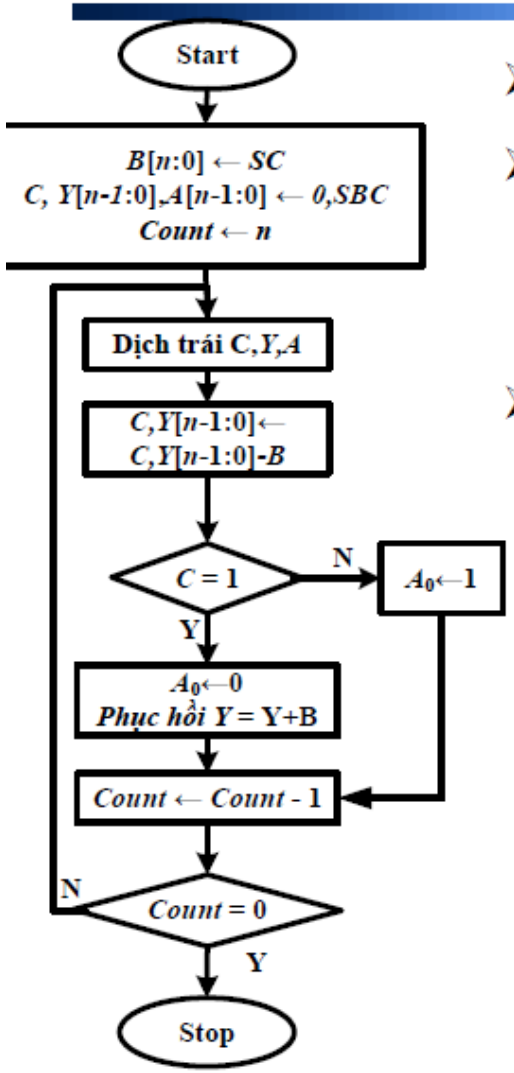
Trừ cho số chia và dịch phải số chia

Ví dụ: Y = A / B

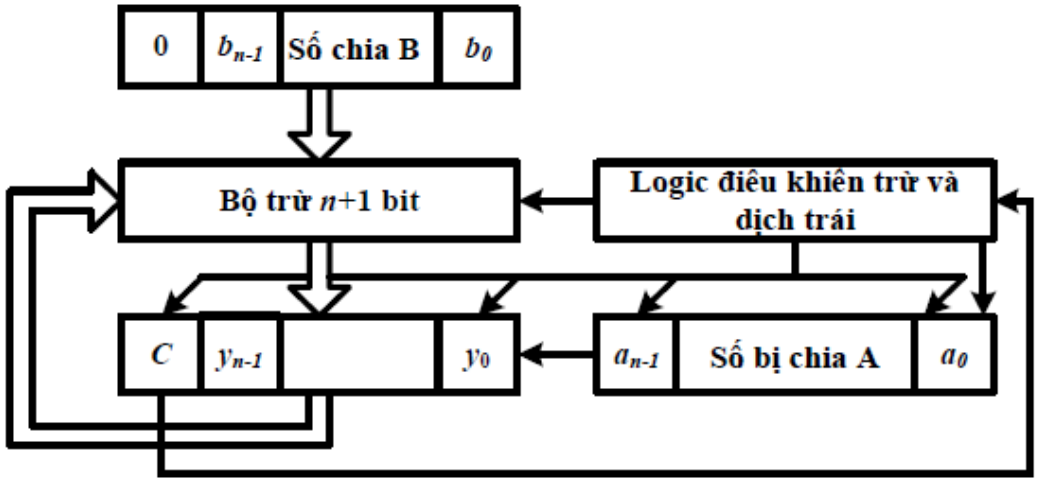


Tối ưu phần cứng: trừ cho số chia và dịch trái phần dư

Thuật toán chia nối tiếp

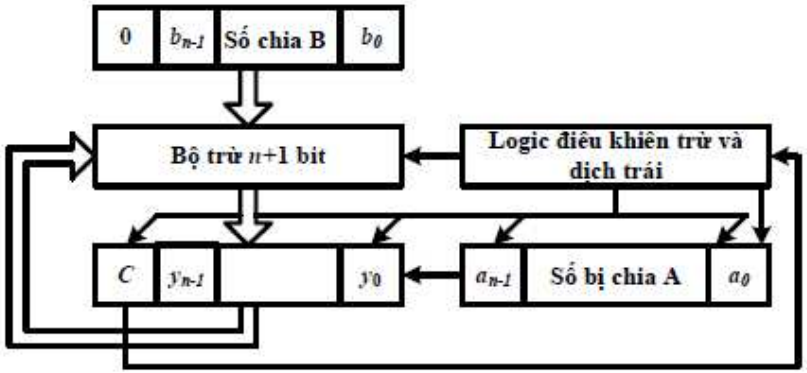


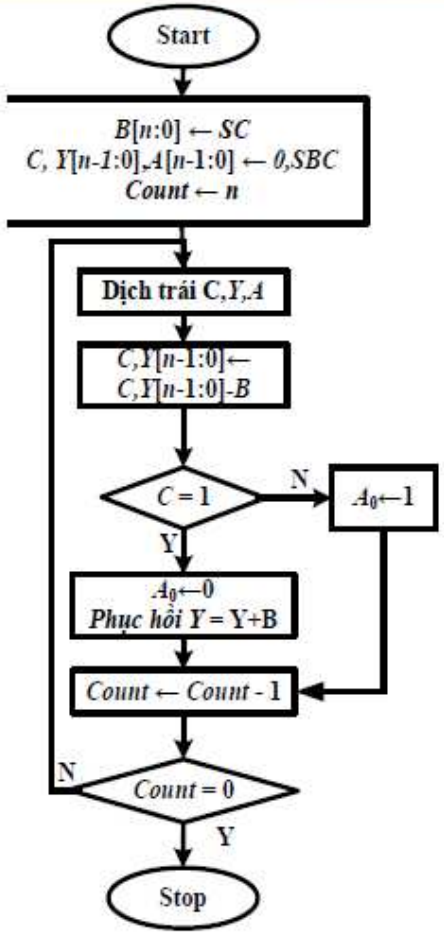
- Trừ Y = Y – B và kiểm tra bit dấu *C* của kết quả
- Nếu *C* = 1 (phép trừ kết quả âm) ➔
  - Bít kết quả a_0=0
  - Phục hồi số bị chia: Y = Y + B
- Nếu *C* = 0 (phép trừ kết quả dương) ➔
  - Bít kết quả a_0=1



Ví dụ – Bộ chia nối tiếp

B		0	0	1	1					
Y	0	0	0	0	0	0	1	1	1	*Counter=4*
Y	0	0	0	0	0	1	1	1	0	
-B	1	1	1	0	1					
Y	1	1	1	0	1	1	1	1	0	
Y	0	0	0	0	0	1	1	1	0	*Counter=3*
Y	0	0	0	0	1	1	1	0	0	
-B	1	1	1	0	1					
Y	1	1	1	1	0	1	1	1	0	
Y	0	0	0	0	1	1	1	0	0	*Counter=2*
Y	0	0	0	1	1	1	0	0	0	
-B	1	1	1	0	1					
Y	0	0	0	0	0	1	0	0	0	
Y	0	0	0	0	0	1	0	0	1	*Counter=1*
Y	0	0	0	0	1	0	0	1	0	
-B	1	1	1	0	1					
Y	1	1	1	1	0	0	0	1	0	
Y	0	0	0	0	1	0	0	1	0	*Counter=0*



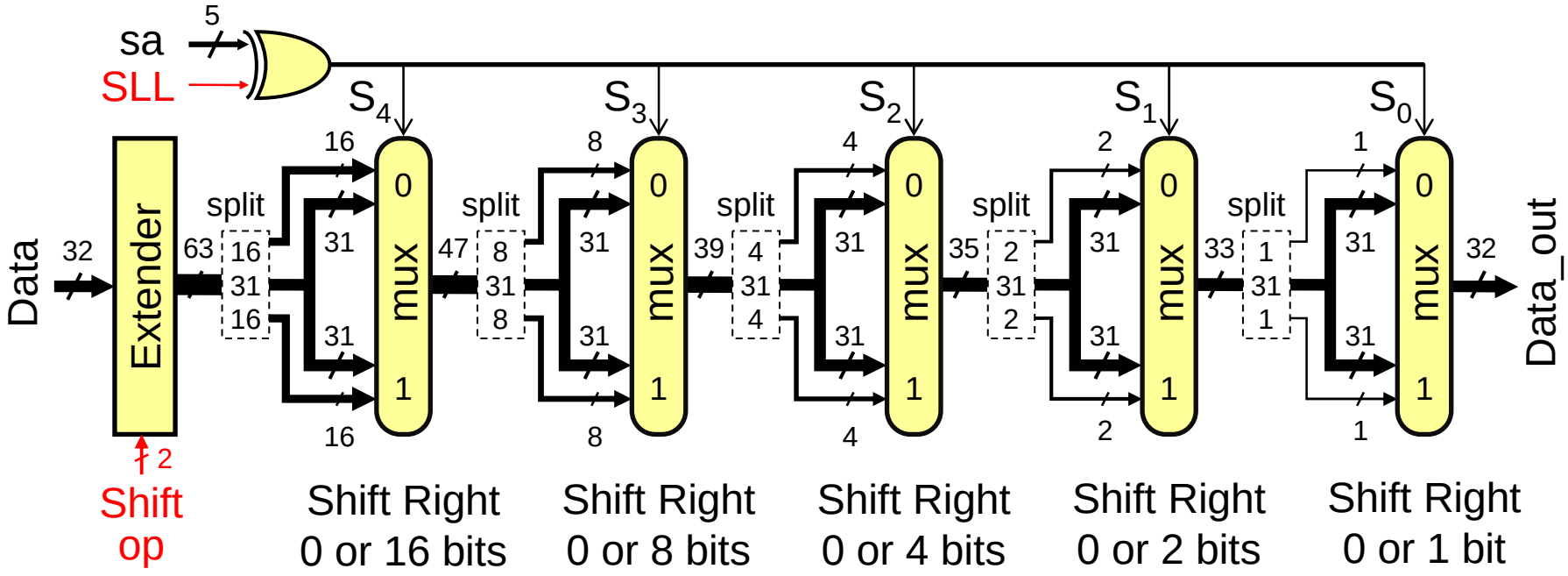


Chia có dấu

1. Chia phần giá trị tuyệt đối
2. Xác định dấu của kết quả
   - Dấu của thương:
     - Dương nếu số chia và số bị chia cùng dấu
     - Âm nếu số chia và số bị chia khác dấu
   - Dấu của phần dư: luôn cùng dấu với số bị chia
3. Đảo kết quả nếu cần thiết

Details of the Shifter

- Implemented with multiplexers and wiring
- Shift Operation can be: SLL, SRL, SRA, or ROR
- Input Data is extended to 63 bits according to Shift Op
- The 63 bits are shifted right according to $S_4S_3S_2S_1S_0$



Details of the Shifter – cont'd

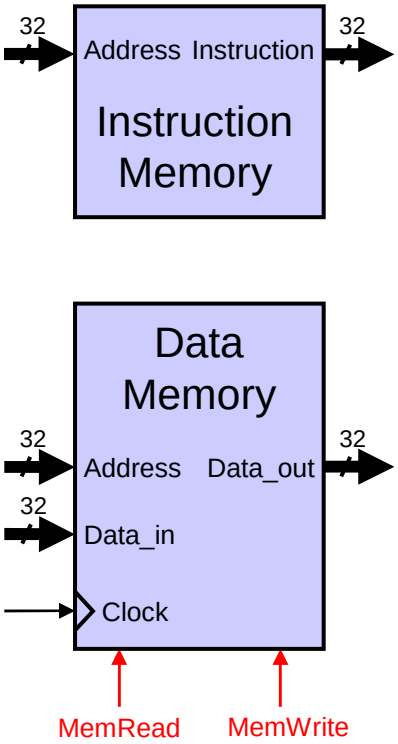
- Input data is extended from 32 to 63 bits as follows:
  - If shift op = SRL then ext_data[62:0] = 0^{31} || data[31:0]
  - If shift op = SRA then ext_data[62:0] = data[31]^{31} || data[31:0]
  - If shift op = ROR then ext_data[62:0] = data[30:0] || data[31:0]
  - If shift op = SLL then ext_data[62:0] = data[31:0] || 0^{31}
- For SRL, the 32-bit input data is zero-extended to 63 bits
- For SRA, the 32-bit input data is sign-extended to 63 bits
- For ROR, 31-bit extension = lower 31 bits of data
- Then, shift right according to the shift amount
- As the extended data is shifted right, the upper bits will be: 0 (SRL), sign-bit (SRA), or lower bits of data (ROR)

Implementing Shift Left Logical

- The wiring of the above shifter dictates a right shift
- However, we can convert a left shift into a right shift
- For SLL, 31 zeros are appended to the right of data
  - To shift left by 0 is equivalent to shifting right by 31
  - To shift left by 1 is equivalent to shifting right by 30
  - To shift left by 31 is equivalent to shifting right by 0
  - Therefore, for SLL use the 1's complement of the shift amount
- ROL is equivalent to ROR if we use (32 – rotate amount)
- ROL by 10 bits is equivalent to ROR by (32–10) = 22 bits
- Therefore, software can convert ROL to ROR

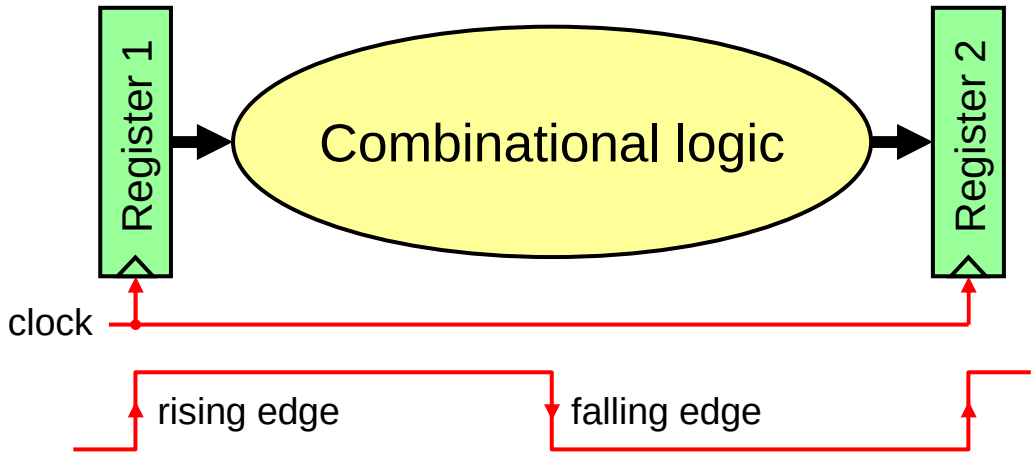
Instruction and Data Memories

- Instruction memory needs only provide read access
  - Because datapath does not write instructions
  - Behaves as combinational logic for read
  - Address selects Instruction after access time
- Data Memory is used for load and store
  - MemRead: enables output on Data_out
    - Address selects the word to put on Data_out
  - MemWrite: enables writing of Data_in
    - Address selects the memory word to be written
    - The Clock synchronizes the write operation
- Separate instruction and data memories
  - Later, we will replace them with caches



Clocking Methodology

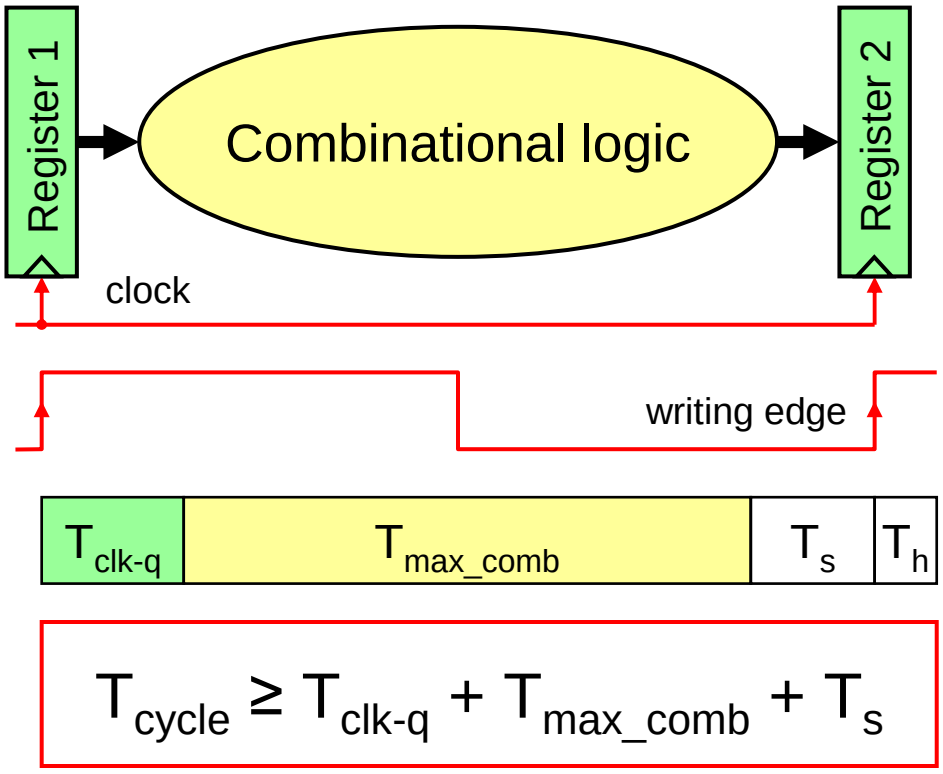
- Clocks are needed in a sequential logic to decide when a state element (register) should be updated
- To ensure correctness, a clocking methodology defines when data can be written and read



- We assume edge-triggered clocking
- All state changes occur on the same clock edge
- Data must be valid and stable before arrival of clock edge
- Edge-triggered clocking allows a register to be read and written during same clock cycle

Determining the Clock Cycle

- With edge-triggered clocking, the clock cycle must be long enough to accommodate the path from one register through the combinational logic to another register



- T_{clk-q} : clock to output delay through register
- T_{max_comb} : longest delay through combinational logic
- T_s : setup time that input to a register must be stable before arrival of clock edge
- T_h: hold time that input to a register must hold after arrival of clock edge
- Hold time (T_h) is normally satisfied since $T_{clk-q} > T_h$

Clock Skew

- Clock skew arises because the clock signal uses different paths with slightly different delays to reach state elements
- Clock skew is the difference in absolute time between when two storage elements see a clock edge
- With a clock skew, the clock cycle time is increased

$$T_{cycle} \geq T_{clk\text{-}q} + T_{max\_combinational} + T_{setup} + T_{skew}$$

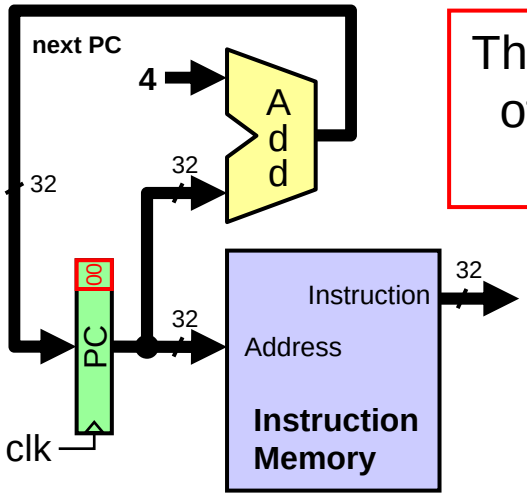
- Clock skew is reduced by balancing the clock delays

Next . . .

- Designing a Processor: Step-by-Step
- Datapath Components and Clocking
- Assembling an Adequate Datapath
- Controlling the Execution of Instructions
- The Main Controller and ALU Controller
- Drawback of the single-cycle processor design

Instruction Fetching Datapath

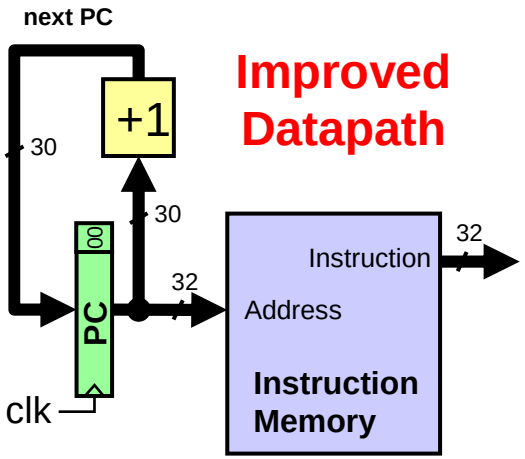
- We can now assemble the datapath from its components
- For instruction fetching, we need …
  - Program Counter (PC) register
  - Instruction Memory
  - Adder for incrementing PC



The least significant 2 bits of the PC are '00' since PC is a multiple of 4

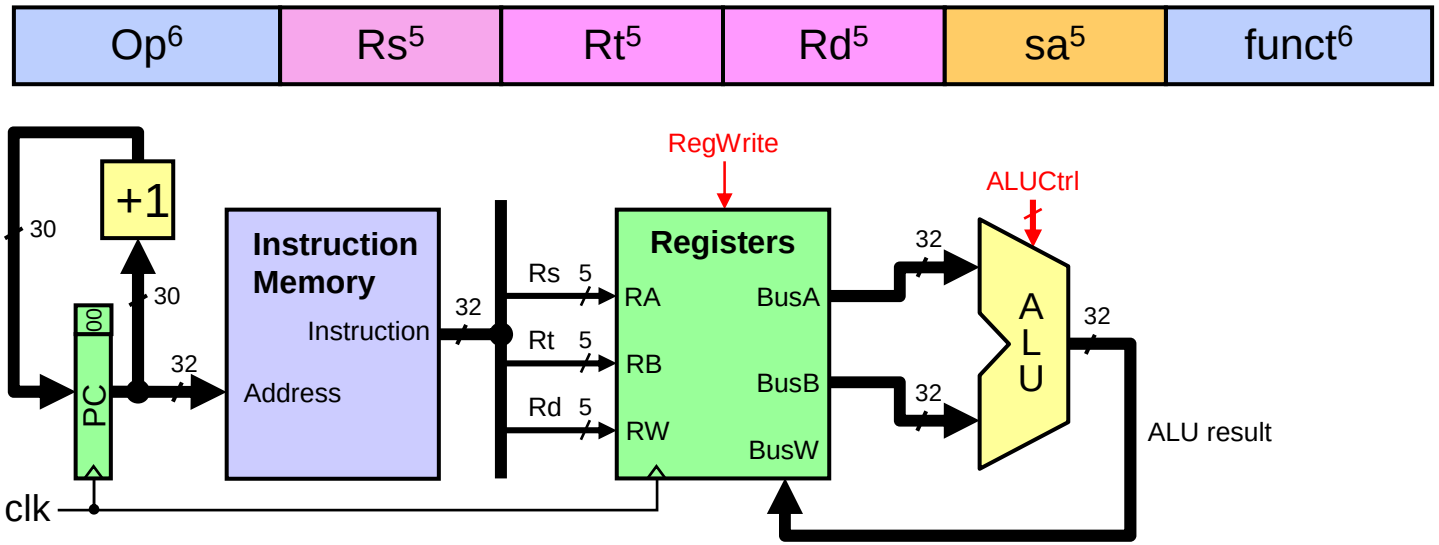
Datapath does not handle branch or jump instructions

Improved datapath increments upper 30 bits of PC by 1



Improved Datapath

Datapath for R-type Instructions



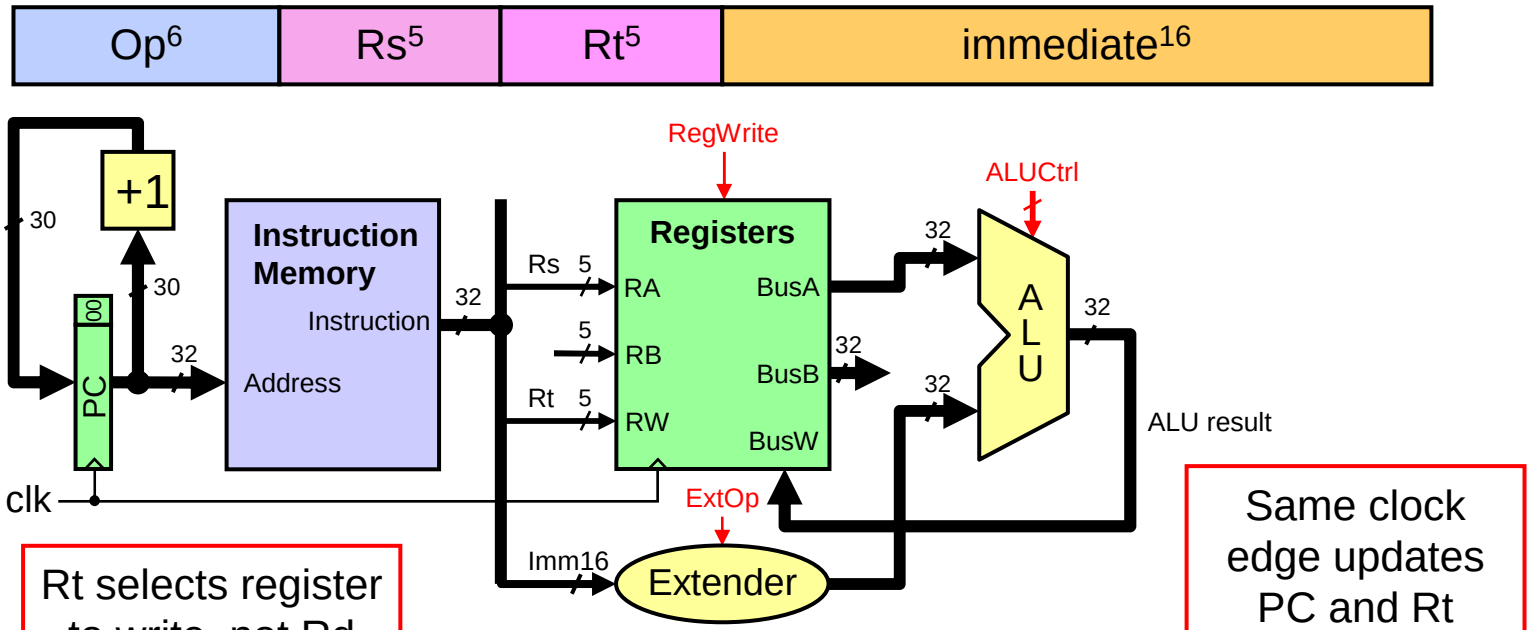
Rs and Rt fields select two registers to read. Rd field selects register to write

BusA & BusB provide data input to ALU. ALU result is connected to BusW

Same clock updates PC and Rd register

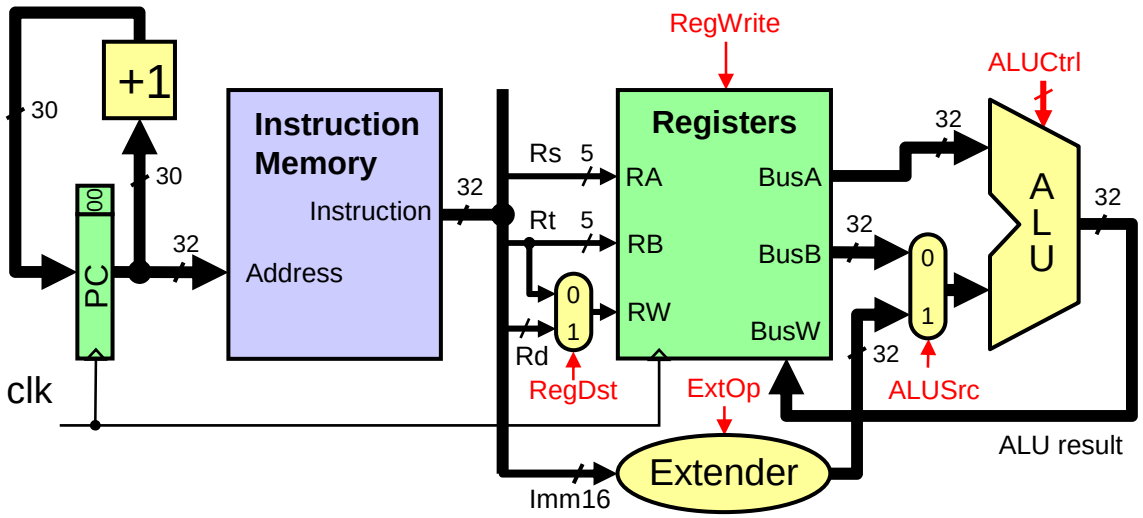
- Control signals
  - ALUCtrl is derived from the funct field because Op = 0 for R-type
  - RegWrite is used to enable the writing of the ALU result

Datapath for I-type ALU Instructions



- Control signals
  - ALUCtrl is derived from the Op field
  - RegWrite is used to enable the writing of the ALU result
  - ExtOp is used to control the extension of the 16-bit immediate

Combining R-type & I-type Datapaths

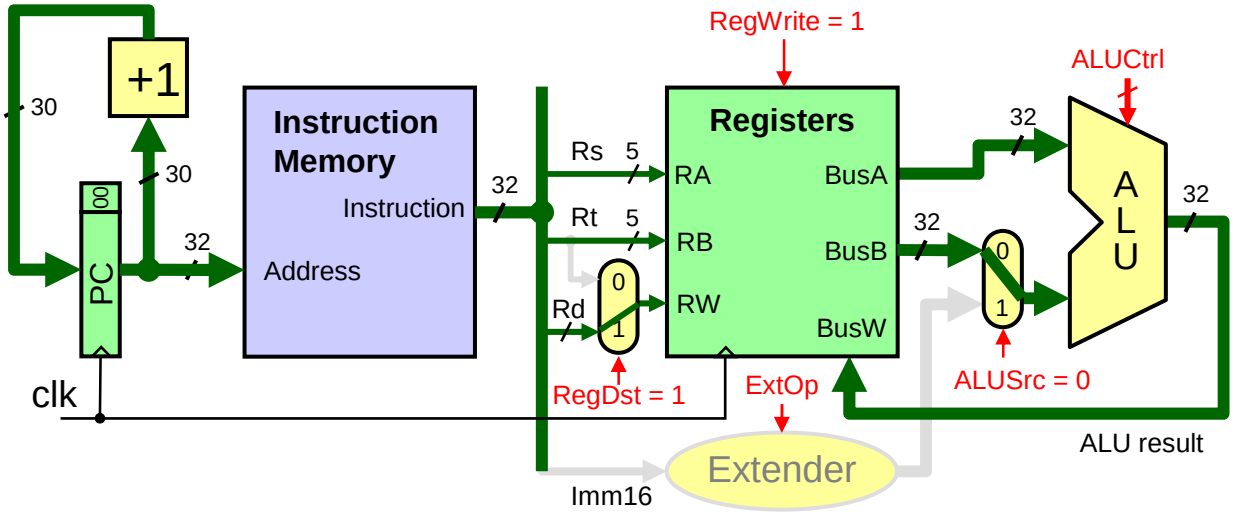


A mux selects RW as either Rt or Rd

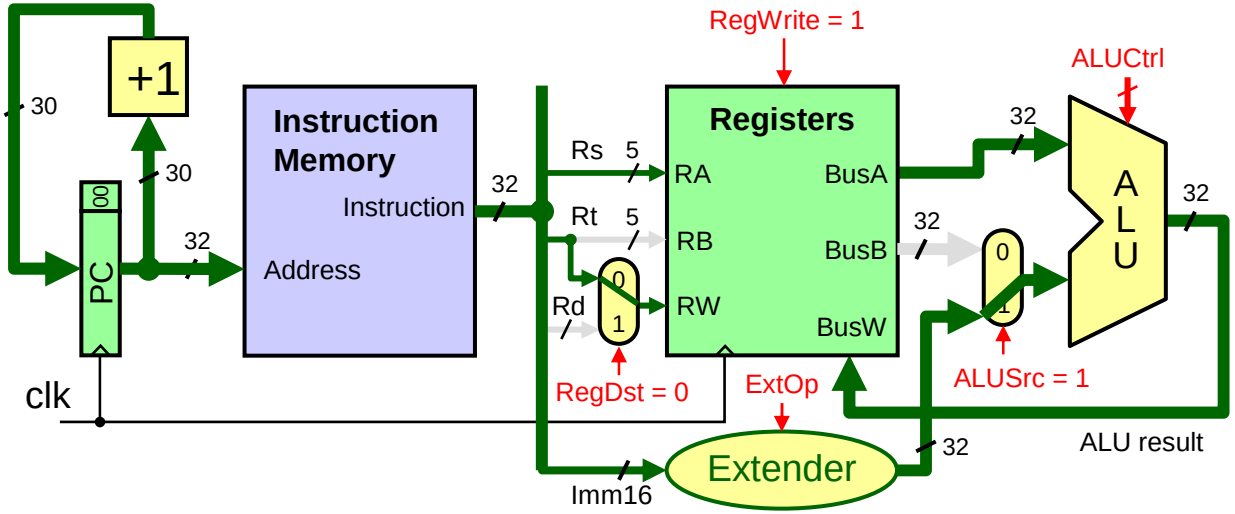
Another mux selects 2nd ALU input as either data on BusB or the extended immediate

- Control signals
  - ALUCtrl is derived from either the Op or the funct field
  - RegWrite enables the writing of the ALU result
  - ExtOp controls the extension of the 16-bit immediate
  - RegDst selects the register destination as either Rt or Rd
  - ALUSrc selects the 2nd ALU source as BusB or extended immediate

Controlling ALU Instructions



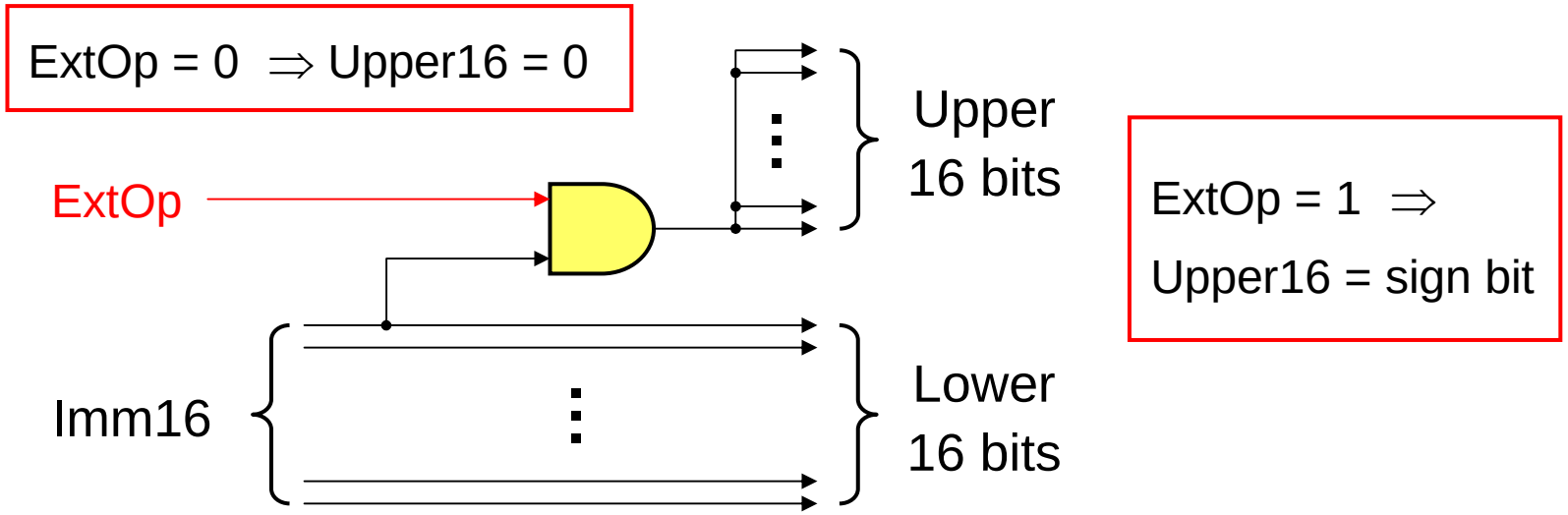
For R-type ALU instructions, RegDst is '1' to select Rd on RW and ALUSrc is '0' to select BusB as second ALU input. The active part of datapath is shown in **green**



For I-type ALU instructions, RegDst is '0' to select Rt on RW and ALUSrc is '1' to select Extended immediate as second ALU input. The active part of datapath is shown in **green**

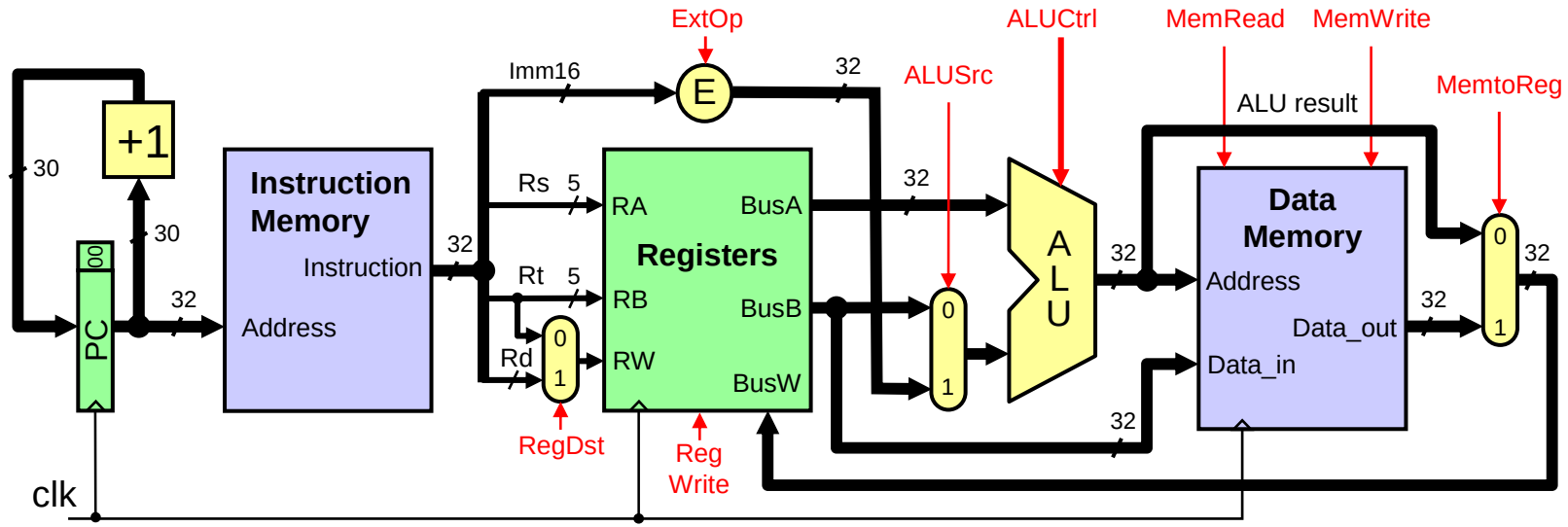
Details of the Extender

- Two types of extensions
  - Zero-extension for unsigned constants
  - Sign-extension for signed constants
- Control signal ExtOp indicates type of extension
- Extender Implementation: wiring and one AND gate



Adding Data Memory to Datapath

- A data memory is added for load and store instructions



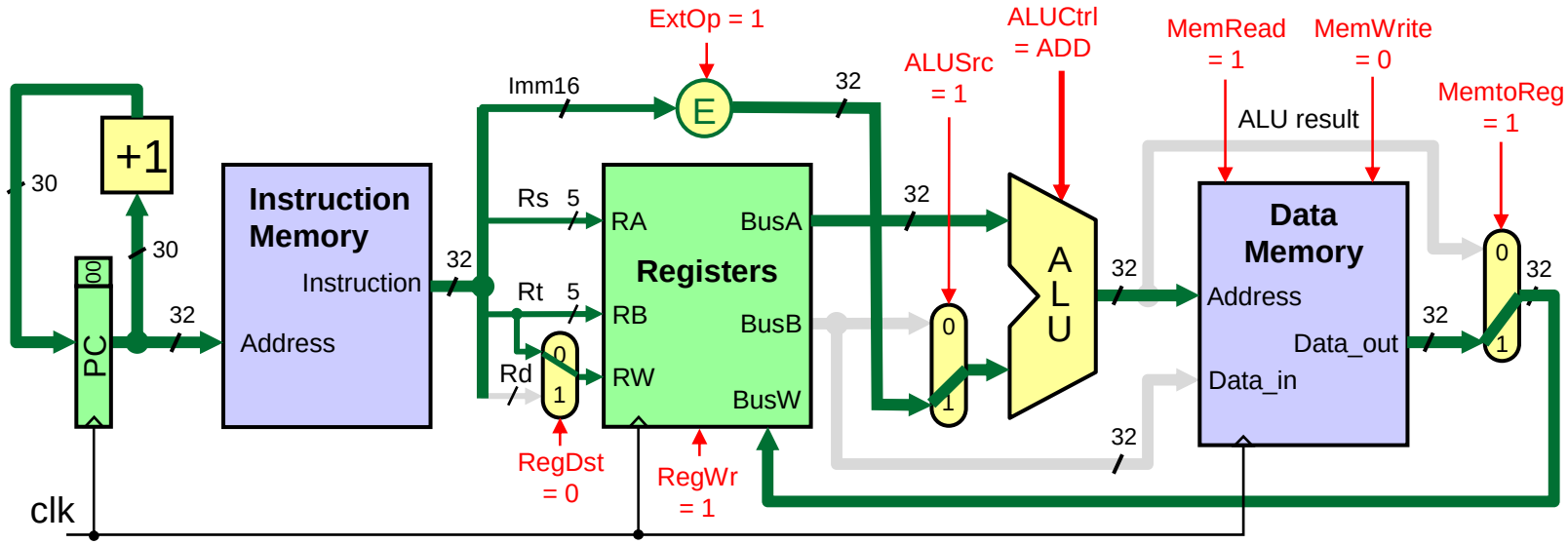
ALU calculates data memory address

A 3rd mux selects data on BusW as either ALU result or memory data_out

- Additional Control signals
  - MemRead for load instructions
  - MemWrite for store instructions
  - MemtoReg selects data on BusW as ALU result or Memory Data_out

BusB is connected to Data_in of Data Memory for store instructions

Controlling the Execution of Load



RegDst = ‘0’ selects Rt as destination register

RegWrite = ‘1’ to enable writing of register file

ExtOp = 1 to sign-extend Immmediate16 to 32 bits

ALUSrc = ‘1’ selects extended immediate as second ALU input

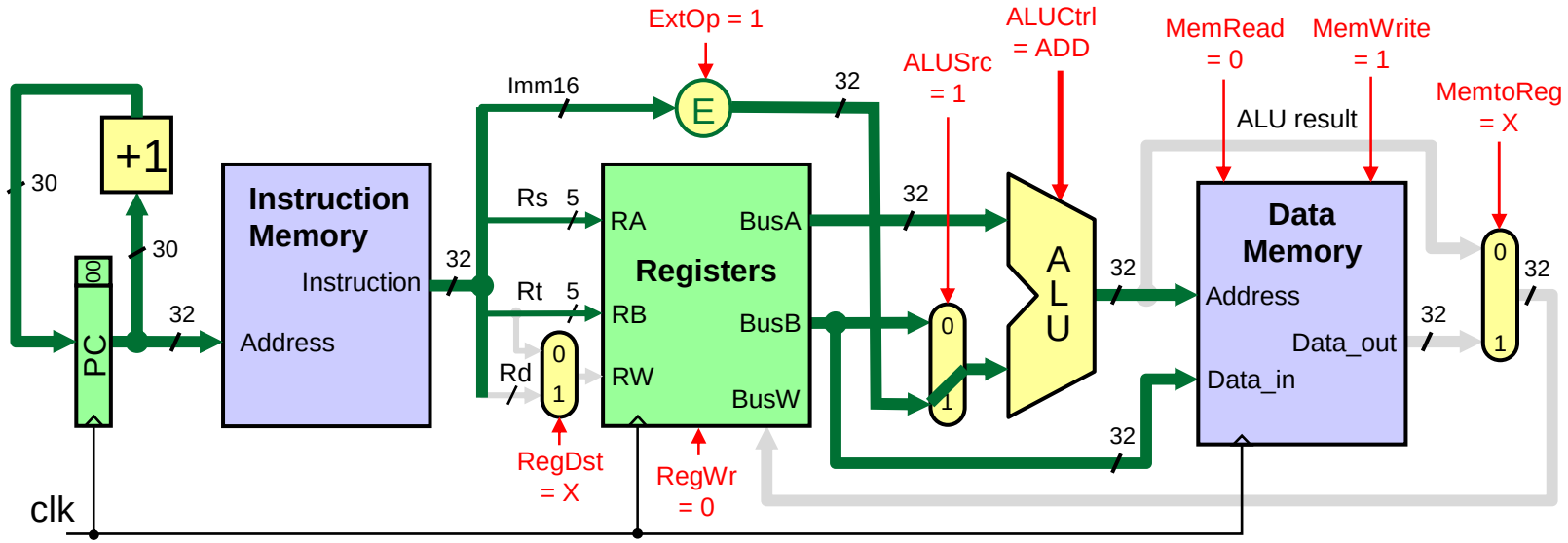
ALUCtrl = ‘ADD’ to calculate data memory address as Reg(Rs) + sign-extend(Imm16)

MemRead = ‘1’ to read data memory

MemtoReg = ‘1’ places the data read from memory on BusW

Clock edge updates PC and Register Rt

Controlling the Execution of Store



RegDst = ‘X’ because no register is written

RegWrite = ‘0’ to disable writing of register file

ExtOp = 1 to sign-extend Immmediate16 to 32 bits

ALUSrc = ‘1’ selects extended immediate as second ALU input

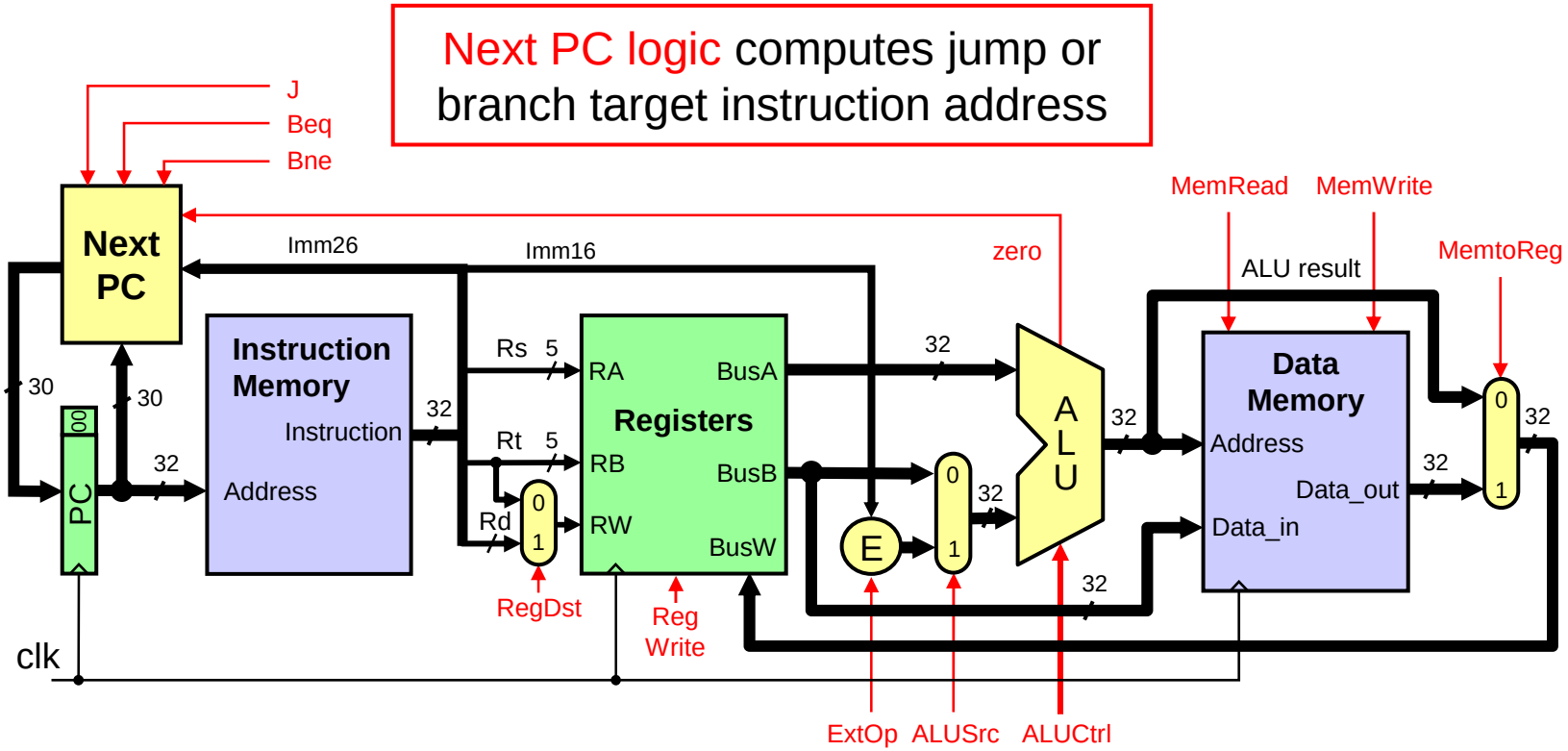
ALUCtrl = ‘ADD’ to calculate data memory address as Reg(Rs) + sign-extend(Imm16)

MemWrite = ‘1’ to write data memory

MemtoReg = ‘X’ because don’t care what data is put on BusW

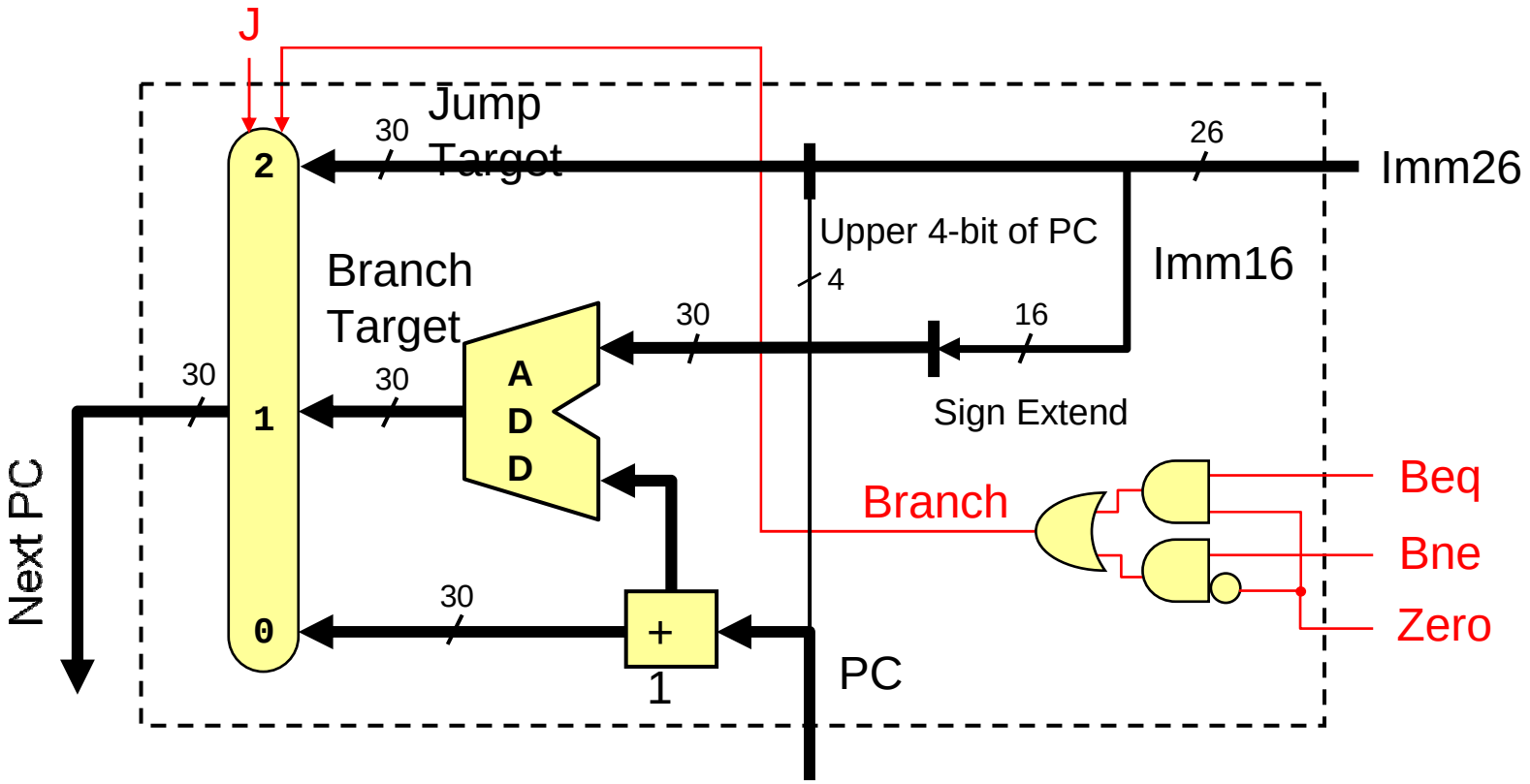
Clock edge updates PC and Data Memory

Adding Jump and Branch to Datapath



- Additional Control Signals
  - J, Beq, Bne for jump and branch instructions
  - Zero condition of the ALU to determine whether branch is taken

Details of Next PC

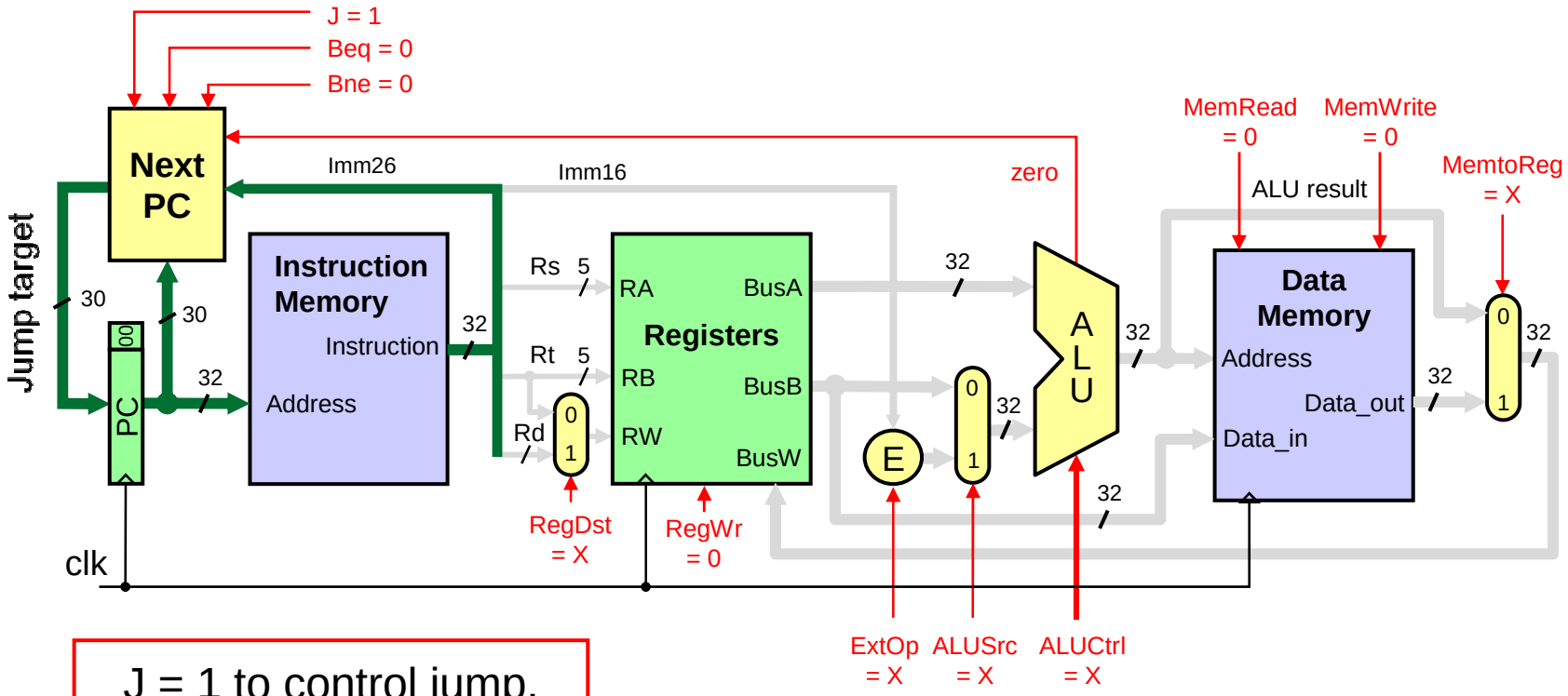


Branch = (Beq . Zero) + (Bne . $\overline{\text{Zero}}$)

Imm16 is sign-extended to 30 bits

Jump Target Address: Upper 4 bits of PC are concatenated with Imm26

Controlling the Execution of Jump



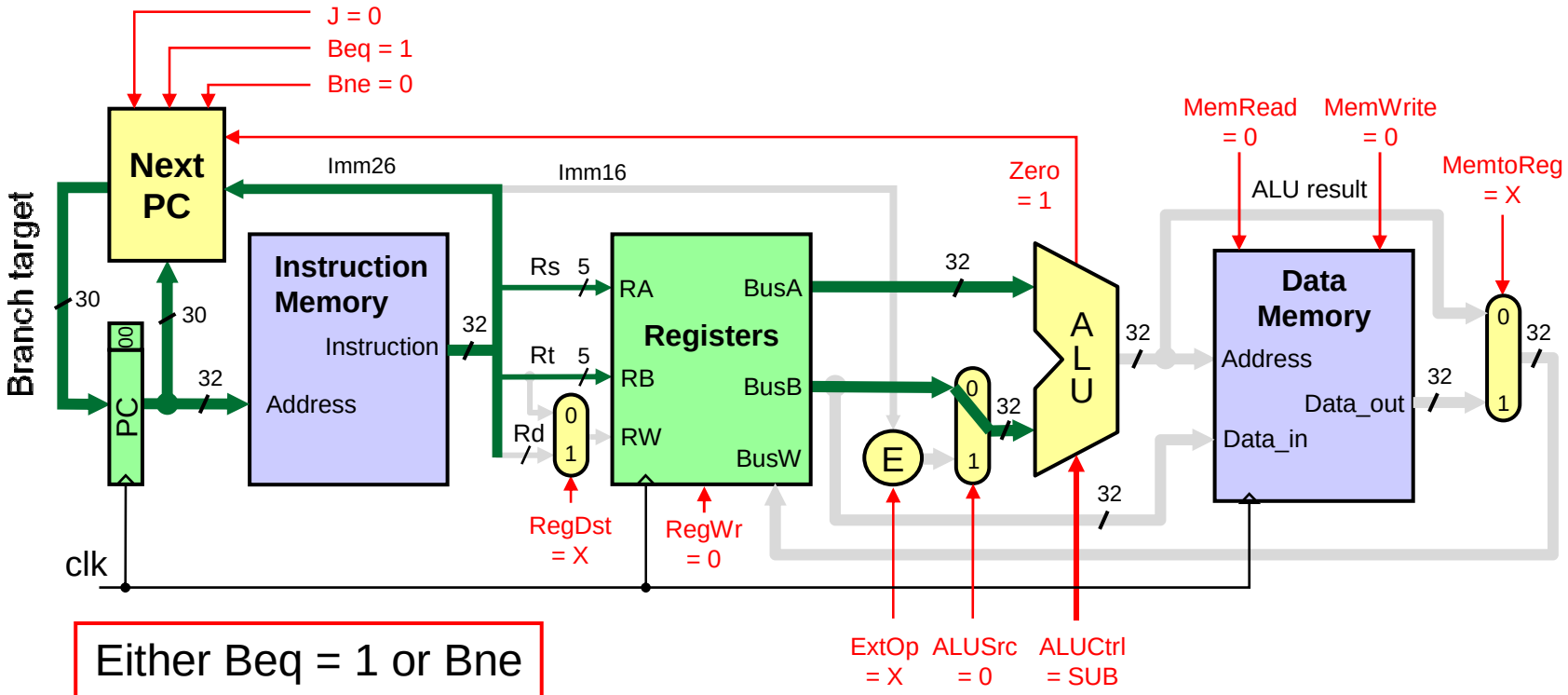
J = 1 to control jump. Next PC outputs Jump Target Address

MemRead, MemWrite, and RegWrite are 0

We don't care about RegDst, ExtOp, ALUSrc, ALUCtrl, and MemtoReg

Clock edge updates PC register only

Controlling the Execution of Branch



Either Beq = 1 or Bne depending on opcode	Next PC outputs branch target if branch is taken
ALUSrc = 0 to select value on BusB	RegWrite, MemRead, and MemWrite are 0
ALUCtrl = SUB to generate Zero Flag	Don't care for RegDst, ExtOp, and MemtoReg
	Clock edge updates PC register only

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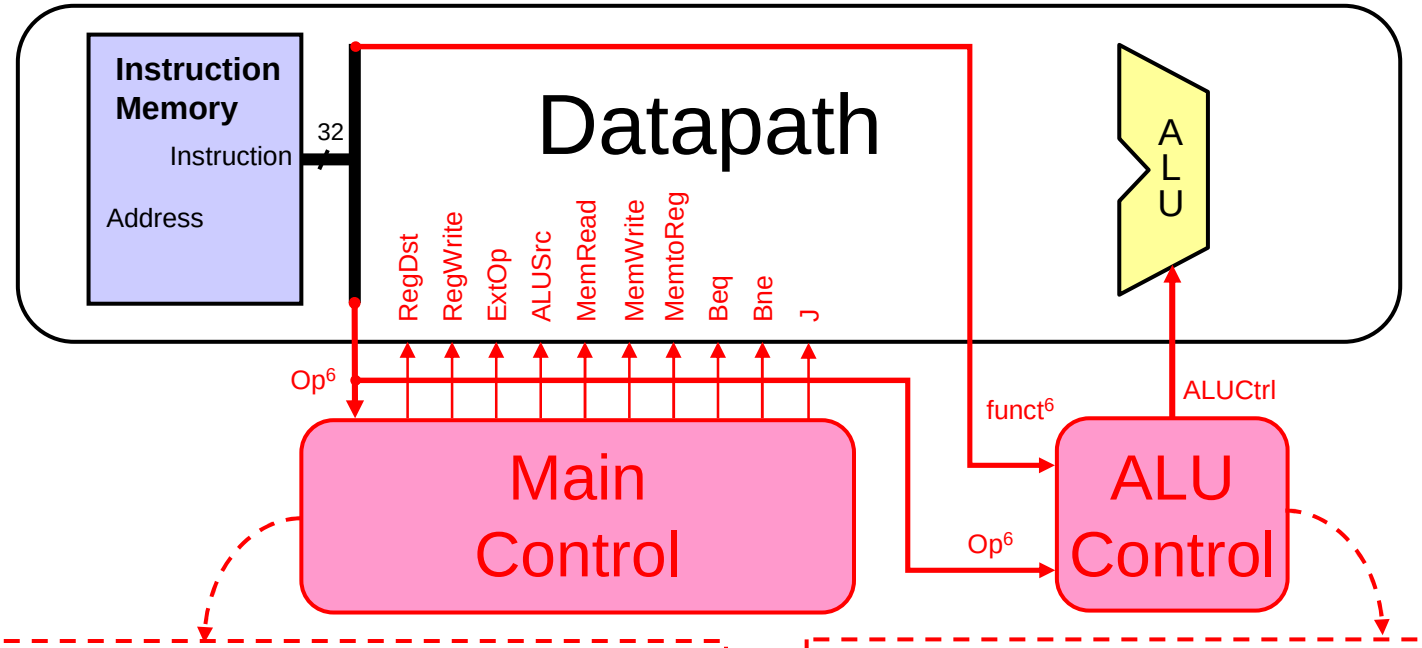
Main Control and ALU Control



Main Control Input:

- 6-bit opcode field from instruction

Main Control Output:

- 10 control signals for the Datapath

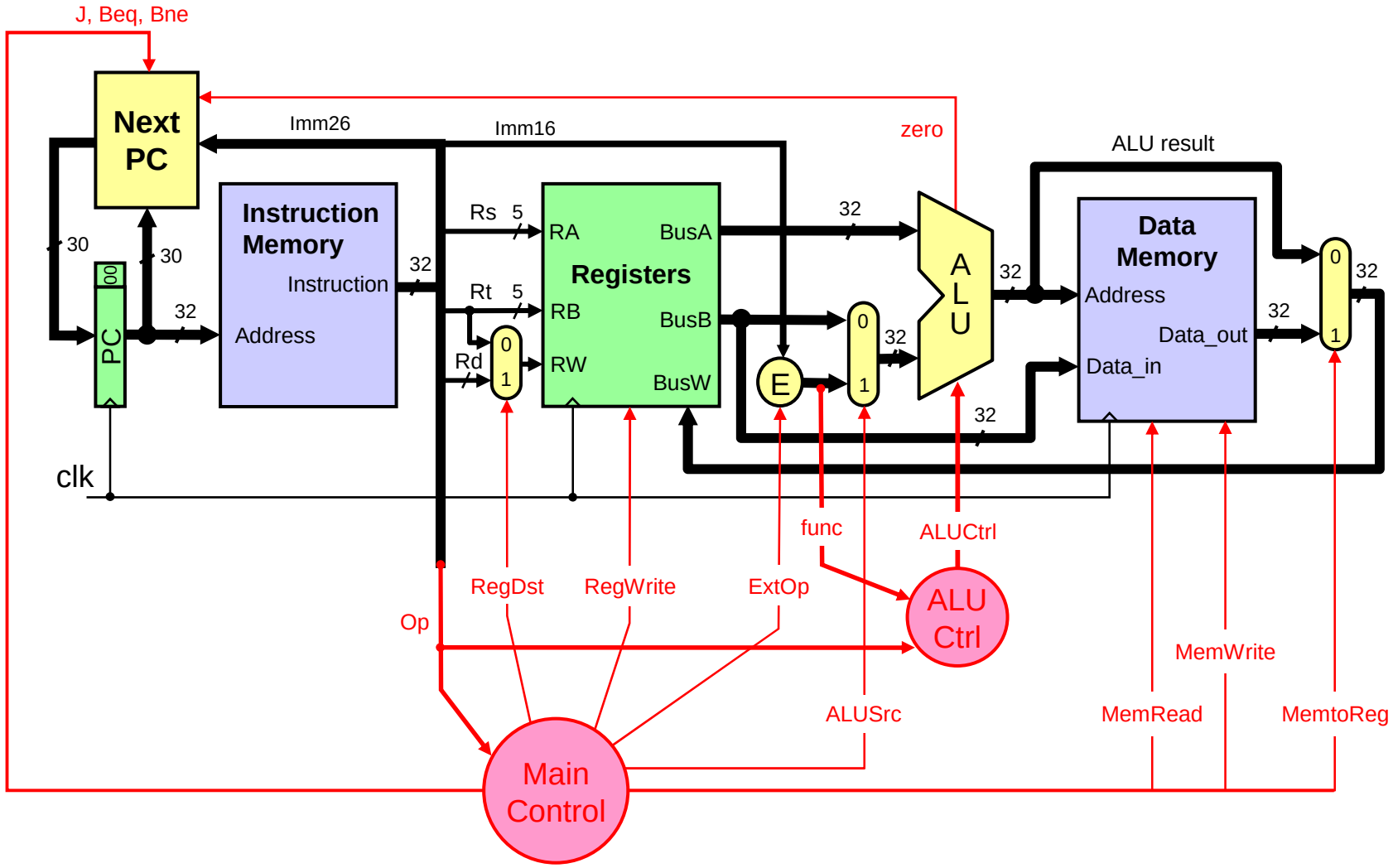
ALU Control Input:

- 6-bit opcode field from instruction
- 6-bit function field from instruction

ALU Control Output:

- ALUCtrl signal for ALU

Single Cycle Datapath + Control



Main Control Signals

Signal	Effect when '0'	Effect when '1'
RegDst	Destination register = Rt	Destination register = Rd
RegWrite	None	Destination register is written with the data value on BusW
ExtOp	16-bit immediate is zero-extended	16-bit immediate is sign-extended
ALUSrc	Second ALU operand comes from the second register file output (BusB)	Second ALU operand comes from the extended 16-bit immediate
MemRead	None	Data memory is read Data_out ← Memory[address]
MemWrite	None	Data memory is written Memory[address] ← Data_in
MemtoReg	BusW = ALU result	BusW = Data_out from Memory
Beq, Bne	PC ← PC + 4	PC ← Branch target address If branch is taken
J	PC ← PC + 4	PC ← Jump target address

Main Control Signal Values

Op	Reg Dst	Reg Write	Ext Op	ALU Src	Beq	Bne	J	Mem Read	Mem Write	Mem toReg
R-type	1 = Rd	1	x	0=BusB	0	0	0	0	0	0
addi	0 = Rt	1	1=sign	1=Imm	0	0	0	0	0	0
slti	0 = Rt	1	1=sign	1=Imm	0	0	0	0	0	0
andi	0 = Rt	1	0=zero	1=Imm	0	0	0	0	0	0
ori	0 = Rt	1	0=zero	1=Imm	0	0	0	0	0	0
xori	0 = Rt	1	0=zero	1=Imm	0	0	0	0	0	0
lw	0 = Rt	1	1=sign	1=Imm	0	0	0	1	0	1
sw	x	0	1=sign	1=Imm	0	0	0	0	1	x
beq	x	0	x	0=BusB	1	0	0	0	0	x
bne	x	0	x	0=BusB	0	1	0	0	0	x
j	x	0	x	x	0	0	1	0	0	x

- X is a don't care (can be 0 or 1), used to minimize logic

Logic Equations for Control Signals

RegDst = R-type

RegWrite = $\overline{(sw + beq + bne + j)}$

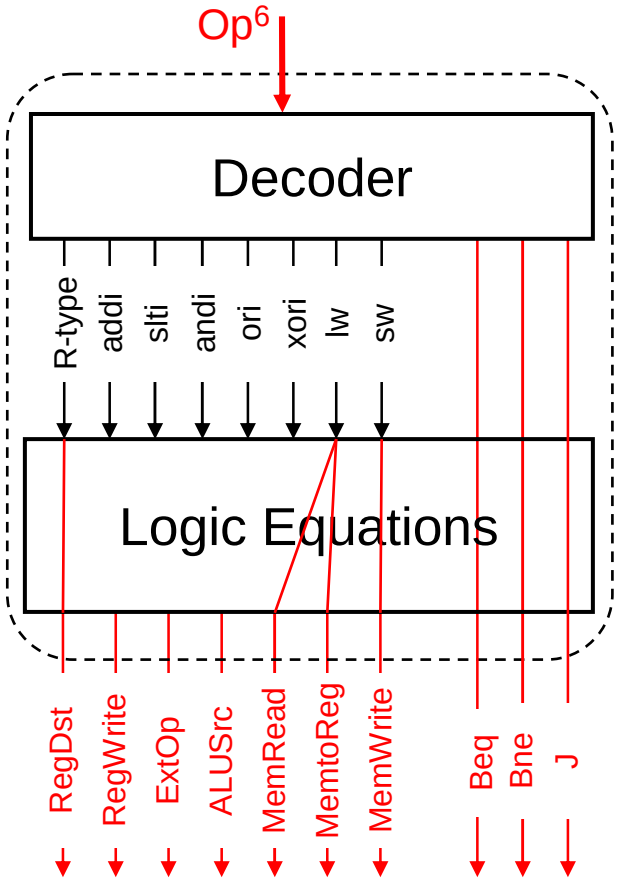
ExtOp = $\overline{(andi + ori + xori)}$

ALUSrc = $\overline{(R\text{-}type + beq + bne)}$

MemRead = lw

MemtoReg = lw

MemWrite = sw



ALU Control Truth Table

Input		Output	4-bit
Op^6	$funct^6$	ALUCtrl	Encoding
R-type	add	ADD	0000
R-type	sub	SUB	0010
R-type	and	AND	0100
R-type	or	OR	0101
R-type	xor	XOR	0110
R-type	slt	SLT	1010
addi	x	ADD	0000
slti	x	SLT	1010
andi	x	AND	0100
ori	x	OR	0101
xori	x	XOR	0110
lw	x	ADD	0000
sw	x	ADD	0000
beq	x	SUB	0010
bne	x	SUB	0010
j	x	x	x

The 4-bit ALUCtrl is encoded according to the ALU implementation

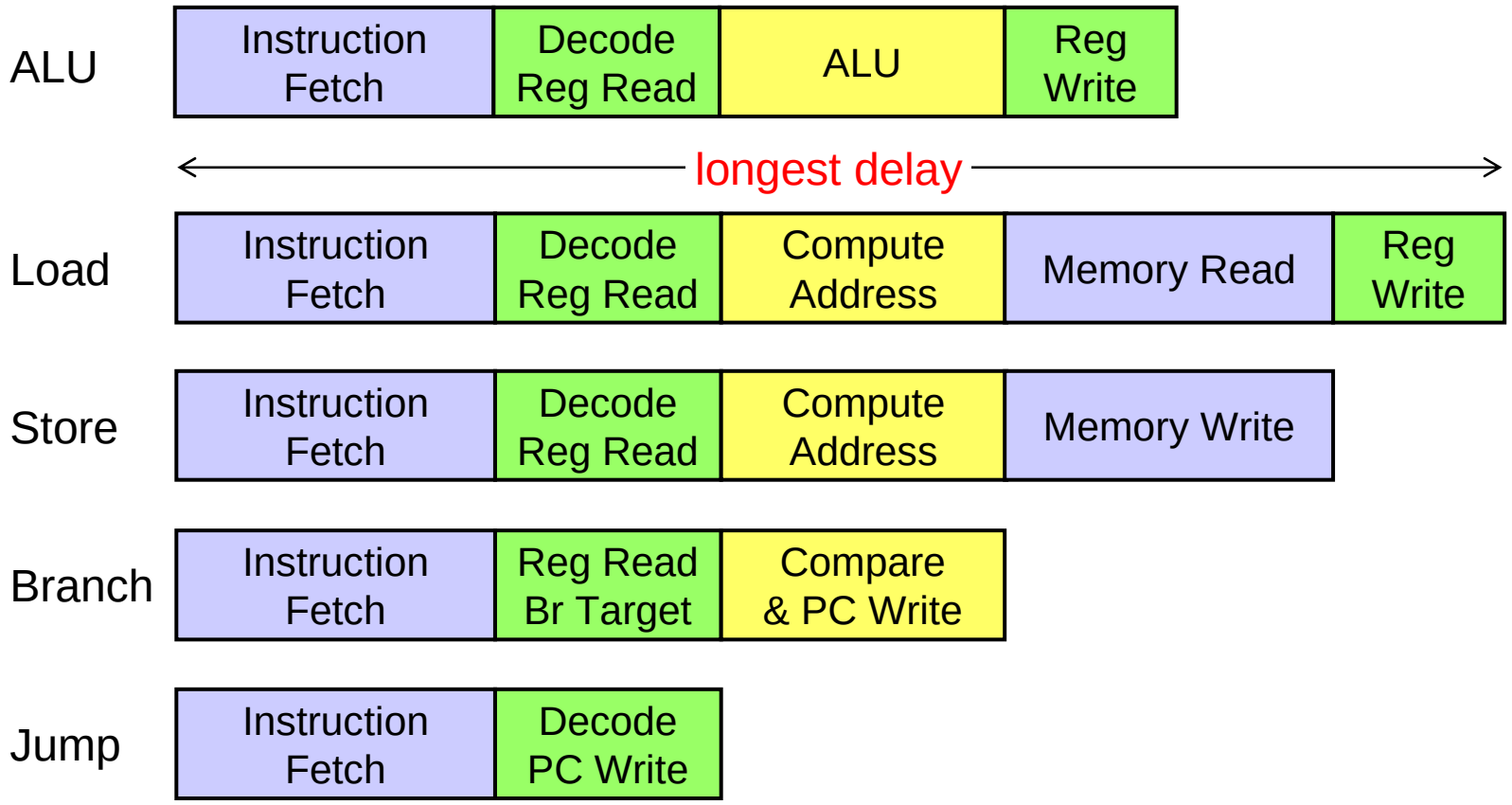
Other ALU control encodings are also possible. The idea is to choose a binary encoding that will simplify the logic

Next . . .

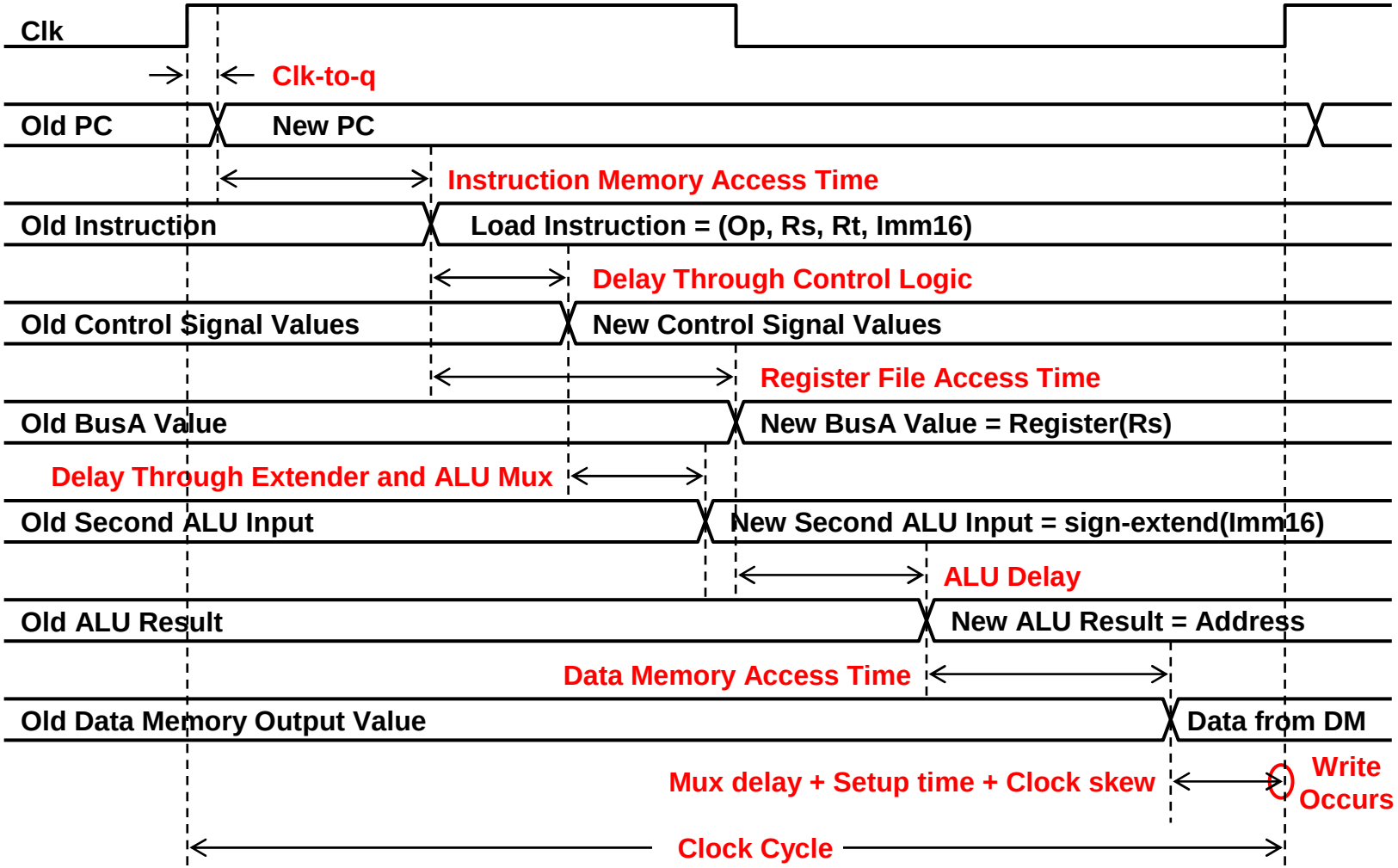
- Designing a Processor: Step-by-Step
- Datapath Components and Clocking
- Assembling an Adequate Datapath
- Controlling the Execution of Instructions
- The Main Controller and ALU Controller
- Drawback of the single-cycle processor design

Drawbacks of Single Cycle Processor

- Long cycle time
  - All instructions take as much time as the slowest instruction



Timing of a Load Instruction



Worst Case Timing – Cont'd

- Long cycle time: long enough for Slowest instruction

  PC Clk-to-Q delay

  + Instruction Memory Access Time

  + Maximum of (

    Register File Access Time,

    Delay through control logic + extender + ALU mux)

  + ALU to Perform a 32-bit Add

  + Data Memory Access Time

  + Delay through MemtoReg Mux

  + Setup Time for Register File Write + Clock Skew

- Cycle time is longer than needed for other instructions
  - Therefore, single cycle processor design is not used in practice

Alternative: Multicycle Implementation

- Break instruction execution into five steps
  - Instruction fetch
  - Instruction decode, register read, target address for jump/branch
  - Execution, memory address calculation, or branch outcome
  - Memory access or ALU instruction completion
  - Load instruction completion
- One clock cycle per step (clock cycle is reduced)
  - First 2 steps are the same for all instructions

Instruction	# cycles	Instruction	# cycles
ALU & Store	4	Branch	3
Load	5	Jump	2

Performance Example

- Assume the following operation times for components:
  - Instruction and data memories: 200 ps
  - ALU and adders: 180 ps
  - Decode and Register file access (read or write): 150 ps
  - Ignore the delays in PC, mux, extender, and wires
- Which of the following would be faster and by how much?
  - Single-cycle implementation for all instructions
  - Multicycle implementation optimized for every class of instructions
- Assume the following instruction mix:
  - 40% ALU, 20% Loads, 10% stores, 20% branches, & 10% jumps

Solution

Instruction Class	Instruction Memory	Register Read	ALU Operation	Data Memory	Register Write	Total
ALU	200	150	180		150	680 ps
Load	200	150	180	200	150	880 ps
Store	200	150	180	200		730 ps
Branch	200	150	180 ← Compare and write PC			530 ps
Jump	200	150 ← Decode and write PC				350 ps

- For fixed single-cycle implementation:
  - Clock cycle = 880 ps determined by longest delay (load instruction)
- For multi-cycle implementation:
  - Clock cycle = max (200, 150, 180) = 200 ps (maximum delay at any step)
  - Average CPI = 0.4×4 + 0.2×5 + 0.1×4+ 0.2×3 + 0.1×2 = 3.8
- Speedup = 880 ps / (3.8 × 200 ps) = 880 / 760 = 1.16

Summary

- 5 steps to design a processor
  - Analyze instruction set => datapath requirements
  - Select datapath components & establish clocking methodology
  - Assemble datapath meeting the requirements
  - Analyze implementation of each instruction to determine control signals
  - Assemble the control logic
- MIPS makes Control easier
  - Instructions are of same size
  - Source registers always in same place
  - Immediates are of same size and same location
  - Operations are always on registers/immediates
- Single cycle datapath => CPI=1, but Long Clock Cycle